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System and method for automated single cell processing

Abstract

A system and method for automated single cell capture and processing is described, where the system includes a deck supporting and positioning a set of sample processing elements; a gantry for actuating tools for interactions with the set of sample processing elements supported by the deck; and a base supporting various processing subsystems and a control subsystems in communication with the processing subsystems. The system can automatically execute workflows associated with single cell processing, including mRNA capture, cDNA synthesis, protein-associated assays, and library preparation, for next generation sequencing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/867,256, filed 5 May 2020, which claims the benefit of U.S. Provisional Application No. 62/844,470, filed on 7 May 2019 and U.S. Application 62/866,726, filed on 26 Jun. 2019, which are each incorporated in its entirety herein by this reference.

TECHNICAL FIELD

(1) This invention relates generally to the cell capture and cell processing field, and more specifically to a new and useful automated system and method for single cell capture and processing in the cell capture and cell processing field.

BACKGROUND

(2) With an increased interest in cell-specific drug testing, diagnosis, and other assays, systems and methods that allow for individual cell isolation, identification, and retrieval are becoming highly desirable. Single cell capture systems and methods have been shown to be particularly advantageous for these applications. However, associated processes and protocols for single cell capture and subsequent analysis often must be performed in a particular order and with a high precision in order to properly maintain the cells. As such, these processes can be time consuming for the user, as well as result in damage to the cells or otherwise unfavorable results if they are not performed properly (e.g., through mistakes in pipetting, through a mix-up of reagents, etc.). In particular, these novel high throughput single cell cytometry assays have great utility in translational medicine, personalized therapy selections, clinical diagnostics, and/or other applications of use, but lack of automation prevents proper performance by novice users, thereby limiting throughput.

(3) Thus, there is a need in the cell capture and cell processing field to create a new and useful system and method for single cell capture and processing.

Description

BRIEF DESCRIPTION OF THE FIGS.

(1) FIGS. 1A-1D depict schematic representations of an embodiment of a system for automated single cell sample processing;

(2) FIGS. 2A-2F depict views of a variation of the system shown in FIGS. 1A-1D;

(3) FIGS. 3A-3C depict variations of a reagent cartridge associated with a system for automated single cell sample processing;

(4) FIGS. 4A-4C depict views of a variation of a sample processing cartridge associated with a system for automated single cell sample processing;

(5) FIGS. 5A-5C depict operation modes of a lid-opening tool associated with the sample processing cartridge shown in FIGS. 4A-4C;

(6) FIGS. 6A-6B depict operation modes of a valve and heating subsystem associated with the sample processing cartridge shown in FIGS. 4A-4C;

(7) FIGS. 7A and 7B depict variations of a tool container and contents associated with a system for automated single cell sample processing;

(8) FIGS. 8A-8I depict variations of features for retaining elements at a deck of a system for

automated single cell sample processing;

(9) FIG. 9 depicts an example of a fluid level detection subsystem of a system for automated single cell sample processing;

(10) FIGS. 10A-10C depict variations of a first subset of components used for material separation in a system for automated single cell sample processing;

(11) FIGS. 11A-11B depict variations of a second subset of components used for material separation in a system for automated single cell sample processing;

(12) FIGS. 12A-12J depict operation modes of a separation subsystem associated with a system for automated single cell sample processing;

(13) FIGS. 13A-13D depict views of components of a variation of a separation subsystem associated with a system for automated single cell sample processing;

(14) FIGS. 14A-14C depict embodiments of functional coupling between components of a system for automated single cell sample processing;

(15) FIG. 15 depicts a flow chart of an embodiment of a method for automated single cell sample processing;

(16) FIG. 16 depicts a first variation of a method for automated single cell sample processing;

(17) FIG. 17 depicts a second variation of a method for automated single cell sample processing;

(18) FIG. 18 depicts a third variation of a method for automated single cell sample processing; and

(19) FIG. 19 depicts a third variation of a method for automated single cell sample processing.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(20) The following description of the preferred embodiments of the invention is not intended to limit the invention to these preferred embodiments, but rather to enable any person skilled in the art to make and use this invention.

(21) 1. Benefits

(22) The invention(s) can confer several benefits over conventional systems and methods.

(23) In particular, the invention(s) confer(s) the benefit of enabling at least partial automation of the protocols involved in single cell capture and subsequent processing, thereby optimizing run success and consistency. In more detail, the user can be removed from part or all of the method (e.g. loading samples, capping lids, on-instrument lysis, reverse transcription processes, cDNA amplification, bead or cDNA product retrieval, on-instrument library preparation and cleanup, etc.). Further, the system and/or method can enable better accuracy of a protocol over conventional systems and methods (e.g. better accuracy in the addition of the correct reagents, better temperature control of reagents, rapid processing of critical liquid handling steps, precise incubation times, optimal bead washing and separation, automated bar code reading, etc.). Further, the system and/or method can confer the benefit of preventing accidents (e.g. knocking the system, spills of reagents, contamination of sample or instrument, etc.), which can commonly occur during the manual performance of a protocol.

(24) Additionally, through use of limited-use and/or pre-loaded and unitized reagent cartridges, the system and/or method can confer the benefit of providing a streamlined user experience with optimized quality control and design architecture to accommodate on-going development of assays and future applications. As such, the system confers the benefit of independent or nearly independent control of reagents or reagent groups. In a specific example of this variation, the system includes a reagent cartridge having any or all of the following dedicated regions: a room temperature region, a cooling region, a heating region, a magnetic region (e.g., overlapping with a heating region), waste capture region, intermediate reagent parking region or any other suitable region. In a related benefit, the system and/or method can confer the benefit of enabling the user to purchase smaller volumes of reagents, such as through the distribution of reagents in protocol-specific types and quantities to be used in accordance with specific automated protocols. This can function to save costs, reduce reagent waste, or have any other suitable outcome.

(25) Additionally, through use of fluid handling and separation elements (e.g., magnetic separation

components), the system and/or method can confer the benefit of providing automated sample and library cleanup steps. Relatedly, the system and/or method can confer the benefit of establishing better fluid flow throughout the system. In a first example, this is enabled through an automated pipetting system (e.g., pipettor, gantry, and assorted pipette tips), which can monitor and/or direct fluid flow (e.g., to maintain an optimal flow rate, to establish an optimal volume of reagents, etc.) without user intervention. The fluid handling system components for single cell preparation and/or other assays may involve use of both of (a) liquid pipettor coupled to a gantry for fluidic dispensing and pumping into a fluidic channel or fluidic reservoir (e.g., of a sample processing cartridge) and/or (b) a built-in on-chip pressurizable waste chamber connected and controlled through a valve integrated with the fluidic network, as described in more detail below. Such a combined dual liquid handling system gives unprecedented control of the flow (e.g., microliter per second to tens of milliliters per second), delivery (e.g., 1-100,000 microliters), and residence time (e.g., milliseconds to hours) of reagents through the fluidic system. Additionally or alternatively, the system can monitor and/or direct fluid flow with user intervention (e.g., with minimal user intervention, to encourage optimal user intervention, etc.).

(26) Additionally, through software and workflow improvements, the system and/or method can minimize number of manual operations performed by a user, and provide relevant system status reports to ensure smooth operation and sample processing.

(27) Additionally, in relation to sample processing disposables, the system and/or method can confer the benefit of consolidating multiple components in a manner that is scalable for disposables having a higher number of sample processing chambers. Additionally, the system can confer the benefit of consolidating two or more conventionally separate processing platform components into a single unit, which can reduce an overall size of the system (e.g., enable a benchtop model), reduce an overall footprint of a mechanism of the system (e.g., pipettor gantry), enable a more efficient transfer of materials among the system, or perform any other suitable function. In a specific example of this variation, an inlet, set of microwells, outlet valve, lid mechanism (e.g., lid of the lid mechanism, full lid mechanism, etc.) and waste chamber are all localized to a single piece.

(28) Additionally, the invention(s) address needs in low parameter flow applications, high parameter flow applications, mass cytometry applications, proteogenomic applications, single cell RNA applications, protein detection applications, single cell multi-omic applications and other applications, by allowing standard users with various skill levels (e.g., novices, experts) to operate platform components. Specific workflows implemented by embodiments of the system are described in more detail below.

(29) In relation to performance, the system and/or method can process cells to generate purified libraries within a day, perform next generation sequencing (NGS) preparation, and perform other processes in a streamlined process (e.g., by a set of dedicated consumables including an efficiently loaded reagent cartridge, a sample processing cartridge, and a container of fluid handling disposables).

(30) Additionally, the system confers the benefit of three-dimensional mobility of a component, such as a pipettor. In a specific example of this variation, the system includes a gantry providing X-Y-Z mobility for a pipettor, enabling the pipettor to perform a variety of tasks (e.g., piercing foil coverings of reagent tubes, transferring materials among a set of wells, etc.) in an automated fashion.

(31) Additionally or alternatively, the system and/or method can confer any other suitable benefit.

(32) 2. System

(33) As shown in FIGS. 1A-1D, an embodiment of a system **100** for automated single cell capture and processing includes: a deck **110** supporting and positioning a set of sample processing elements; a gantry **170** for actuating tools for interactions with the set of sample processing elements supported by the deck **110**; and a base **180** supporting various processing subsystems and a control

subsystems in communication with the processing subsystems, wherein the control subsystems control states of the deck **110**, the set of sample processing elements, and the gantry **170** in order to transition the system **100** between various operation modes. Embodiments, variations, and examples of operation modes, which provide various workflows, are described in further detail in Section 3 below.

(34) Embodiments of the system **100** function to enable automated single cell capture and any or all of associated processing of the captured cells. In more detail, the user can be removed from part or all of the method (e.g. loading samples, capping lids, on-instrument lysis, reverse transcription processes, cDNA amplification, bead or cDNA product retrieval, on-instrument library preparation and cleanup, etc.). The system can additionally or alternatively function to enhance the accuracy (e.g. by minimizing manual processes) of cell capture and sample processing protocols.

Additionally, through use of limited-use and/or pre-loaded reagent cartridges, the system **100** can provide a streamlined user experience with optimized quality control and design architecture to accommodate on-going development of assays and future applications. As such, the system confers the benefit of independent or nearly independent control of reagents or reagent groups. In a specific example of this variation, the system includes a reagent cassette having any or all of the following dedicated regions: a room temperature region, a cooling region, a heating/thermo-cycling region, a magnetic region (e.g., overlapping with a heating region), a region to provide a cell sample input, a region for prepared library output or any other suitable region. In a related benefit, the system and/or method can confer the benefit of enabling the user to purchase smaller volumes of reagents, such as through the distribution of reagents in protocol-specific types and quantities to be used in accordance with specific automated protocols. This can function to save costs, reduce reagent waste, or have any other suitable outcome.

(35) Additionally, through use of fluid handling and separation elements (e.g., magnetic separation components), embodiments of the system **100** can function to provide automated sample and library cleanup steps. Relatedly, the system **100** can confer the benefit of establishing better fluid flow throughout the system. In a first example, this is enabled through an automated pipetting system (e.g., pipettor, gantry, and assorted pipette tips), which can monitor and/or direct fluid flow (e.g., to maintain an optimal flow rate, to establish an optimal volume of reagents, etc.) without or with minimal user intervention.

(36) Additionally, the system **100** can enable low parameter flow applications, high parameter flow applications, mass cytometry applications, proteogenomic applications, single cell RNA applications, protein detection applications, and other applications, by allowing standard users with various skill levels (e.g., novices, experts) to operate platform components. Furthermore, in relation to performance, the system **100** can process cells or other biological material to rapidly generate purified libraries, perform next generation sequencing (NGS) preparation, and perform other processes in a streamlined process (e.g., by a set of dedicated consumables including an efficiently loaded reagent cartridge, a sample processing cartridge, and a container of fluid handling disposables).

(37) In specific embodiments, the system **100** can comply with use requirements including one or more of: providing automated processes for nucleic acid library preparation, ability to provide quality control at desired points of a run, providing complete and single use kits for various assays, providing validated and locked protocols, providing alignment and retention of various system components, providing means for monitoring and controlling system operation (e.g., with a touch display), providing remote monitoring capabilities, providing sample processing within 24 hours, providing visual and/or audible system notifications, providing the ability to be cleaned with standard laboratory cleaners and without disassembly, fitting on a standard laboratory bench, providing easy installation, providing assay materials with stable shelf life, returning reports of maintenance history, providing data storage (e.g., in relation to external storage media, in relation to cloud storage, etc.), providing training, and providing other suitable functions according to

various requirements.

(38) As described above, in relation to sample processing, embodiments of the system **100** can include or be configured to process cells, cell-derived material, and/or other biological material (e.g., cell-free nucleic acids). The cells can include any or all of mammalian cells (e.g., human cells, mouse cells, etc.), embryos, stem cells, plant cells, or any other suitable kind of cells. The cells can contain target material (e.g., target lysate, mRNA, RNA, DNA, etc.) which originates within the cells and is optionally captured by the cell capture system for processing. Additionally, the containers containing the cells can be prepared from multiple cell-containing samples (e.g., 12 samples, 24 samples, 48 samples, 96 samples, 384 samples, 1536 samples, other numbers of samples), wherein the various samples are hashed or barcoded prior to mixing them together into a single container (or reduced number of containers). This feature enables automated processing of multiple samples in the same automated run for their respective single cell preparation and library preparation operations. Additionally or alternatively, the system **100** can be configured to interact with particles (e.g., beads, probes, nucleotides, oligonucleotides, polynucleotides, etc.), droplets, encapsulated cells, encapsulated biomarkers, reagents, or any other suitable materials.

(39) The system can further additionally or alternatively include any or all of the system components as described in U.S. application Ser. No. 16/048,104, filed 27 Jul. 2018; U.S. application Ser. No. 16/049,057, filed 30 Jul. 2018; U.S. application Ser. No. 15/720,194, filed 29 Sep. 2017; U.S. application Ser. No. 15/430,833, filed 13 Feb. 2017; U.S. application Ser. No. 15/821,329, filed 22 Nov. 2017; U.S. application Ser. No. 15/782,270, filed 12 Oct. 2017; U.S. application Ser. No. 16/049,240, filed 30 Jul. 2018; U.S. application Ser. No. 15/815,532, filed 16 Nov. 2017; U.S. application Ser. No. 16/115,370, filed 28 Aug. 2018, U.S. application Ser. No. 16/564,375, filed 9 Sep. 2019, and U.S. application Ser. No. 16/816,817, filed 12 Mar. 2020, which are each incorporated in their entirety by this reference.

(40) 2.1 System: Deck

(41) As shown in FIGS. **1A-1D**, the deck **110** functions as a platform to support and position one or more components of the system **100** (e.g., at a top broad surface, at a top and bottom broad surface, at a side surface, etc.) for automated sample processing. Furthermore, the deck **110** can function to position one or more components of the system **100** to align with or otherwise interact with fluid processing subsystems, heating subsystems, separation subsystems (e.g., magnetic separation subsystems), and/or other subsystems coupled to the gantry **170** and/or base **180**, as described below. In this regard, the deck **110** can be stationary as a reference platform, while other components are actuated into position for interacting with elements of the deck **110**. Alternatively, the deck **110** can be coupled to one or more actuators for positioning elements of the deck **110** for interactions with other subsystems.

(42) In the embodiment shown in FIGS. **1A-1D**, the deck **110** provides a platform supporting the set of sample processing elements, where the sample processing elements can include disposable and/or reusable components, where the components include containers for containing sample processing materials and/or tools for processing samples (e.g., in relation to fluid handling, in relation to material separation, in relation to heating and cooling, etc.). In embodiments, the deck **110** can support a set of sample processing elements including one or more units of: a reagent cartridge **120**, a sample processing cartridge **130**, a tool container **140**, a heating and cooling subsystem **150**, a pumping subsystem **157**, a fluid level detection subsystem **159**, and a separation subsystem **160**. Additionally or alternatively, the deck **110** can include other suitable components (e.g., fluorescence detection subsystems, confocal microscope subsystems, spectroscopic detection subsystems, Total Internal Reflection Fluorescence (TIRF) subsystems, Nuclear Magnetic Resonance (NMR) subsystems, Raman Spectroscopy (RS) RS subsystems, etc.).

(43) The sample processing elements can be supported in a co-planar manner by the deck **110**, or alternatively at different planes. Preferably, discrete elements supported by the deck are non-overlapping, but alternative embodiments of the deck **110** can support the sample processing

elements in an overlapping manner (e.g., for conservation of space, etc., for operational efficiency, etc.).

(44) As shown in FIGS. 1A and 1D, the deck **110** can be accessible by door **90**, where the door **90** of the system **100** can transition between open and/or closed modes in order to provide access to the deck **110** and elements supported by the deck **110**. However, in other variations, the deck **110** may not be accessible by door **90**.

(45) Details of embodiments, variations, and examples of elements supported by the deck **110** are further described in Sections 2.1.1 through 2.1.5 below.

(46) 2.1.1 Deck-Supported Element: Reagent cartridge

(47) The deck **110** includes at least one region **111** (shown in FIGS. 2A and 2B) for supporting a unit of the reagent cartridge **120**, where the region **111** functions to position the reagent cartridge **120** relative to portions of the heating and cooling subsystem **150**, and separation subsystem **160** described in more detail below. In this regard, the region **111** can include one or more openings, recesses, and/or protrusions for providing interfaces between complementary portions of the reagent cartridge **120** and associated portions of the heating and cooling subsystem **150** and separation subsystem **160**, and additionally to promote and maintain alignment between such portions.

(48) The reagent cartridge **120** functions to contain, in one or more compartments, materials for cell capture and/or processing of samples according to one or more workflows for various applications. As such, the reagent cartridge **120** can define a set of storage volumes distributed across a set of domains, where the set of domains can be configured for providing suitable environments for the material contents of each domain. The set of storage volumes can directly contain sample processing materials, and/or can alternatively be configured to receive and maintain positions of individual containers (e.g., tubes, etc.) that contain sample processing materials. The storage volumes of each domain can be distributed in arrays, or otherwise arranged. Storage volumes can have circular cross sections, rectangular cross sections, or other morphologies (e.g., cross sections, widths, depths, etc.) depending upon application of use (e.g., cold storage, heat transfer, magnetic separation, etc.).

(49) The set of domains can additionally or alternatively be configured to provide modularity, where one or more domains can be pre-packaged with materials that are stable over longer shelf lives, while other domains can be configured to receive materials that have short shelf lives (e.g., immediately prior to use). The set of domains can additionally or alternatively be configured to promote operational efficiency (e.g., in relation to grouping similar materials, etc.) for apparatuses that interact with materials of the reagent cartridge **120**. The set of domains can additionally or alternatively define regions for receiving and/or processing material (e.g., nucleic acid material) extracted from the sample processing cartridge **130** described in more detail below.

(50) Additionally or alternatively, domains of the set of domains can be separate (e.g. domain for receiving heat is separate from domains that are intended for other storage temperatures or applications requiring different temperatures), overlapping, or otherwise arranged. Domains of the set of domains can additionally or alternatively be distinguished from each other by a morphology (e.g., length of the storage volumes of each domain, depth of storage volumes for accessing or interfacing with other elements of the deck, width or depth of domains configured for efficient heat transfer, etc.). In some variations, the set of domains can further include at least one domain supporting an absorbent or porous material pad that can be used for receiving drips of fluid (e.g., from a tip of a pipettor, described below) during processing. The internal surface properties for certain domains (e.g., for PCR reactions, for magnetic separation, etc.) may be configured with high surface polish to enable low binding or retention of biomolecules (e.g., nucleic acids or proteins). The various domains may also be mixed and matched to provide a large number of available assays to the customers.

(51) Individual storage volumes of the set of storage volumes of the reagent cartridge **120** can

further include one or more seals, which function to isolate materials within the reagent cartridge **120**, to prevent cross-contamination between materials within individual storage volumes, to prevent contaminants from entering individual storage volumes, and/or to prevent evaporative loss during storage and shipment. The seal(s) can be puncturable seal(s) (e.g., composed of paper, composed of a metal foil, and/or composed of any other suitable material). However, the seal(s) can alternatively be configured to be non-puncturable (e.g., the seal(s) can be configured to peel away from the reagent cartridge **120**). In embodiments, certain reagent containers may also be sealed by a hinged lid that can be opened or closed by a tool (e.g., as described in more detail below), as needed for processing at appropriate steps of the protocol.

(52) In variations, the set of domains can include a first domain for storing reagents requiring a chilled environment (e.g., at a temperature from 1C-15C), a second domain for storing materials that can be stored in ambient conditions, a third domain storing tubes with materials for performing polymerase chain reaction (PCR) operations and interfacing with heating elements described below, a fourth domain for storing functionalized particles (e.g., beads with probes having barcoding regions and other functional regions, as described in U.S. application Ser. No. 16/115,370, etc.), and a fifth domain for performing separation operations (e.g., separation of target from non-target material by magnetic force). In variations, domains providing different environments for the storage volumes can be configured differently. For instance, the first domain (i.e., for cold storage) can be composed of a thermally insulating material and/or can include insulating material about storage volumes of the domain (e.g., individually, about the entire domain). Additionally or alternatively, a domain for separation can be include magnetically conductive materials configured to provide proper magnetic field characteristics for separation. Additionally or alternatively, domains for thermocycling or other heat transfer applications can be configured with thermally conductive materials to promote efficient heat transfer to and from the reagent cartridge. In embodiments, various domains can be optimally positioned such that there is minimal cross-talk between certain operations. For example, the domain(s) for chilled reagent storage volumes can be maintained a temperature (e.g., 4C) during a run, whereas the domain(s) for PCR reactions can require heating (e.g., up to 95C during denature). As such, to minimize the effect of PCR thermocycling on chilled reagents, the domain(s) containing the reagents stored at ambient temperature may be configured in between the PCR thermocycling domain(s) and chilled domain(s). In order to further prevent heat cross-talk, additional buffer tubes with just air may be used in between critical domains that need independent temperature control.

(53) In variations, process materials supported by the domains of the reagent cartridge **120** can include one or more of: buffers (e.g. ethanol, priming buffer, lysis buffer, custom lysis buffers, sample wash buffers, saline with RNAase inhibitors, bead wash buffers, RT buffer, buffer, etc.), oils (e.g. perfluorinert oil), PCR master mixtures, cells, beads (e.g. functionalized beads) or any other suitable materials used for cell capture and/or sample processing. Additionally or alternatively, one or more of the set of storage volumes can be empty (e.g. initially empty, empty throughout one or more processes, empty prior to filling by an operator, etc.). Different storage regions in various domains of the reagent cartridge can have initial reagent volumes from a few microliters (e.g., 5 microliters) to 50 milliliters. Additional details of process materials and applications of use are described below in relation to workflows of Section 3.

(54) In a specific example, as shown in FIG. 3A, the reagent cartridge **120'** includes a first domain **121'** at a first peripheral region of the reagent cartridge **120'** for storing reagents requiring a chilled environment, a second domain **122'** at a central region of the reagent cartridge **120'** for storing materials that can be stored in ambient conditions, a third domain **123'** at a peripheral region of the cartridge, near the second domain **122**, for storing tubes with materials for performing polymerase chain reaction (PCR) operations, a fourth domain **124'** at a peripheral region of the reagent cartridge **120'**, for storing functionalized particles (e.g., beads with probes having barcoding regions and other functional regions, as described in U.S. application Ser. No. 16/115,370, etc.),

and a fifth domain **125** at a peripheral region of the reagent cartridge **120'** for performing separation operations (e.g., separation of target from non-target material by magnetic force). In the specific example, the fourth domain **124'** can be a modular element, whereby the fourth domain **124'** can be stored separately from the rest of the reagent cartridge **120'** until the functionalized particles are ready for use, at which point the fourth domain **124'** is set in position and coupled with the reagent cartridge **120'**.

(55) In the specific example, the first domain **121'** and the second domain **122'** are covered by a first seal composed of a metal foil, the third domain **123'** and the fifth domain **125'** are covered by a second seal composed of a paper, and the fourth domain **124'** is covered by a third seal composed of a metal foil. However, variations of the example of the reagent cartridge **120'** can be configured in another suitable manner.

(56) In another specific example for a 3.1. RNA processing protocol (e.g., corresponding to the workflow of Section 3.1. below) shown in FIG. 3B, the reagent cartridge **120''** can include: a first storage volume **1201** (e.g., having a volume of 4.46 mL) for 100% molecular grade ethanol; a second storage volume **1202** (e.g., having a volume of 6.6 mL) for a first wash buffer; a third storage volume **1203** (e.g., having a volume of 1.1 mL) for a particle binding buffer; a fourth storage volume **1204** (e.g., having a volume of 1.1 mL) for a lysis buffer; a fifth storage volume **1205** (e.g., having a volume of 1.1 mL) for perfluorinert oil; a sixth storage volume **1206** (e.g., having a volume of 6.63 mL) for a particle binding wash solution; a seventh storage volume **1207** (e.g., having a volume of 1.1 mL) for a pre RT reaction wash buffer; an eighth storage volume **1208** (e.g., having a volume of 1 mL) for a 0.1M sodium hydroxide solution; a ninth storage volume **1209** (e.g., having a volume of 2.5 mL) for a second wash buffer; a tenth storage volume **1210** (e.g., having a volume of 1 mL) for mineral oil; an 11.sup.th storage volume **1211** (e.g., having a volume of 1.1 mL) for 80% molecular grade ethanol; a 12.sup.th storage volume **1212** (e.g., having a volume of 2.2 mL) for nuclease-free water; a 13.sup.th storage volume **1213** (e.g., having a volume of 12.36 mL) for waste; a 14.sup.th storage volume **1214** (e.g., having a volume of 0.15 mL) for 0.1M DTT; a 15.sup.th storage volume **1215** for an RT cocktail without superscript IV; a 16.sup.th storage volume **1216** (e.g., having a volume of 0.011 mL) for superscript IV enzyme; a 17.sup.th storage volume **1217** (e.g., having a volume of 0.22 mL) for exonuclease buffer; an 18.sup.th storage volume **1218** (e.g., having a volume of 0.022 mL) for exonuclease enzyme; a 19.sup.th storage volume **1219** (e.g., having a volume of 0.128 mL) for a second strand synthesis mixture; a 20.sup.th storage volume **1220** (e.g., having a volume of 0.072 mL) for a second strand synthesis primer; a 21.sup.st storage volume **1221** (e.g., having a volume of 0.22 mL) for a PCR master mixture for mRNA amplification; a 22.sup.nd storage volume **1222** (e.g., having a volume of 0.33 mL) for a mixture (e.g., Kapa Biosystems™ HiFi HotStart Ready Mixture (2×)); a 23.sup.rd storage volume **1223** (e.g., having a volume of 0.025 mL) for mRNA product; a 24.sup.th storage volume **1224** (e.g., having a volume of 0.11 mL) for functionalized particles; a 25.sup.th storage volume **1225** (e.g., having a volume of 0.03 mL) for magnetic retrieval particles; a 26.sup.th storage volume **1226** (e.g., having a volume of 0.66 mL) for AMPure XP particles; 27.sup.th storage volume **1227** for magnetic particle retrieval collection; a 28.sup.th storage volume **1228** for magnetic particle preparation; a 29.sup.th storage volume **1229** for magnetic particle second strand synthesis; a 30.sup.th storage volume **1230** for magnetic particle retrieval post-PCR cDNA amplification; a 31.sup.st storage volume **1231** for AMPure XP particle purification; a 32.sup.nd storage volume **1232** for a first exonuclease treatment; a 33.sup.rd storage volume **1233** for a second exonuclease treatment; a 34.sup.th storage volume **1234** for second strand synthesis; a 35.sup.th storage volume **1235** for a first cDNA amplification operation; a 36.sup.th storage volume **1236** for a second cDNA amplification operation; a 37.sup.th storage volume **1237** for a third cDNA amplification operation; a 38.sup.th storage volume **1238** for a fourth cDNA amplification operation; and a 39.sup.th storage volume **1239** for a cell suspension.

(57) In another specific example for a CITE-Seq processing protocol (e.g., corresponding to

workflow in Section 3.3 below) shown in FIG. 3C, the reagent cartridge **120'''** can include: a first storage volume **1201'** (e.g., having a volume of 4.46 mL) for 100% molecular grade ethanol; a second storage volume **1202'** (e.g., having a volume of 6.6 mL) for a first wash buffer; a third storage volume **1203'** (e.g., having a volume of 1.1 mL) for a particle binding buffer; a fourth storage volume **1204'** (e.g., having a volume of 1.1 mL) for a lysis buffer; a fifth storage volume **1205'** (e.g., having a volume of 1.1 mL) for perfluorinert oil; a sixth storage volume **1206'** (e.g., having a volume of 6.63 mL) for a particle binding wash solution; a seventh storage volume **1207'** (e.g., having a volume of 1.1 mL) for a pre RT reaction wash buffer; an eighth storage volume **1208'** (e.g., having a volume of 1 mL) for a 0.1M sodium hydroxide solution; a ninth storage volume **1209'** (e.g., having a volume of 2.5 mL) for a second wash buffer; a tenth storage volume **1210'** (e.g., having a volume of 1 mL) for mineral oil; an 11.sup.th storage volume **1211'** (e.g., having a volume of 1.1 mL) for 80% molecular grade ethanol; a 12.sup.th storage volume **1212'** (e.g., having a volume of 2.2 mL) for nuclease-free water; a 13.sup.th storage volume **1213'** (e.g., having a volume of 12.36 mL) for waste; a 14.sup.th storage volume **1214'** (e.g., having a volume of 0.15 mL) for 0.1M DTT; a 15.sup.th storage volume **1215'** for an RT cocktail without superscript IV; a 16.sup.th storage volume **1216'** (e.g., having a volume of 0.011 mL) for superscript IV enzyme; a 17.sup.th storage volume **1217'** (e.g., having a volume of 0.22 mL) for exonuclease buffer; an 18.sup.th storage volume **1218'** (e.g., having a volume of 0.022 mL) for exonuclease enzyme; a 19.sup.th storage volume **1219'** (e.g., having a volume of 0.128 mL) for a second strand synthesis mixture; a 20.sup.th storage volume **1220'** (e.g., having a volume of 0.072 mL) for a second strand synthesis primer; a 21.sup.st storage volume **1221'** (e.g., having a volume of 0.22 mL) for a PCR master mixture for cDNA amplification; a 22.sup.nd storage volume **1222'** (e.g., having a volume of 0.33 mL) for a mixture (e.g., Kapa Biosystems™ HiFi HotStart Ready Mixture (2×)); a 23.sup.rd storage volume **1223'** for an indexing primer; a 24.sup.th storage volume **1224'** (e.g., having a volume of 0.11 mL) for a PCR master mixture for mRNA amplification; a 25.sup.th storage volume **1225'** (e.g., having a volume of 0.2 mL) for ADT product; a 26.sup.th storage volume **1226'** (e.g., having a volume of 0.025 mL) for mRNA product; a 27.sup.th storage volume **1227'** (e.g., having a volume of 0.11 mL) for functionalized particles; a 28.sup.th storage volume **1228'** (e.g., having a volume of 0.03 mL) for magnetic retrieval particles; a 29.sup.th storage volume **1229'** (e.g., having a volume of 0.66 mL) for AMPure XP particles; 30.sup.th storage volume **1230'** for magnetic particle retrieval collection; a 31.sup.st storage volume **1231'** for magnetic particle preparation; a 32.sup.nd storage volume **1232'** for magnetic particle second strand synthesis; a 33.sup.rd storage volume **1233'** for magnetic particle retrieval post-PCR cDNA amplification; a 34.sup.th storage volume **1234'** for AMPure XP particle purification; a 35.sup.th storage volume **1235'** for a first antibody-derived tag purification operation; a 36.sup.th storage volume **1236'** for a second antibody-derived tag purification operation; a 37.sup.th storage volume **1237'** for AMPure XP particle ADT post-PCR purification; a 38.sup.th storage volume **1238'** for mRNA AMPure XP particle purification; a 39.sup.th storage volume **1239'** for mRNA AMPure XP particles post-PCR purification; a 40.sup.th storage volume **1240'** for a first exonuclease treatment; a 41.sup.st storage volume **1241'** for a second exonuclease treatment; a 42.sup.nd storage volume **1242'** for second strand synthesis; a 43.sup.rd storage volume **1243'** for a first cDNA amplification operation; a 44.sup.th storage volume **1244'** for a second cDNA amplification operation; a 45.sup.th storage volume **1245'** for a third cDNA amplification operation; a 46.sup.th storage volume **1246'** for a fourth cDNA amplification operation; a 47.sup.th storage volume **1247'** for antibody-derived tag fraction amplification; a 48.sup.th storage volume **1248'** for mRNA amplification; and a 49.sup.th storage volume **1249'** for a cell suspension.

(58) 2.1.2 Deck-Supported Element: Sample cartridge

(59) As shown in FIGS. 2A and 2B, the deck **110** also includes at least one region **112** for supporting a unit of the sample processing cartridge **130**, where the region **112** functions to position the sample processing cartridge **130** relative to portions of the heating and cooling subsystem **150**,

the pumping subsystem **157**, and the fluid level detection subsystem **159** described in more detail below. In this regard, the region **112** can include one or more openings, recesses, and/or protrusions for providing interfaces between complementary portions of the sample processing cartridge **130** and associated portions of the heating and cooling subsystem **150**, the pumping subsystem **157**, and the fluid level detection subsystem **159**, and additionally to promote and maintain alignment between such portions.

(60) The sample processing cartridge **130** functions to provide one or more sample processing regions in which cells are captured and optionally sorted, processed, or otherwise treated for downstream applications, where the downstream applications can be performed on the sample processing cartridge **130** (e.g., on-chip) and/or away from the sample processing cartridge **130** (e.g., off-chip). Portions of the sample processing cartridge **130** can be configured within a single substrate, but can additionally or alternatively include multiple portions (e.g. connected by fluidic pathways) across multiple substrates.

(61) As shown in FIGS. 4A-4C, an example of the sample processing cartridge **130'** can include a base substrate **131** to which other elements are coupled and/or in which other elements are defined. Furthermore, in relation to sample processing involving microfluidic elements, the base substrate **131** can function as a manifold for fluid transfer to microfluidic elements, accessing of sample processing volumes at various stages of processing, and transfer of waste materials produced during sample processing. In variations, the base substrate **131** supports one or more of: a sample processing chip **132**, an inlet reservoir **133** for receiving sample material (e.g., containing cells, containing particles, etc.) and delivering it into the sample processing chip **132**, an access region **134** for accessing one or more regions of the sample processing chip **132**, a lid **135** covering the access region and including a gasket **136** providing sealing functions, and a waste containment region **137** for receiving waste material from the sample processing chip **132**. The cartridge may have additional gasketed ports to also connect with off-cartridge pumping system present in the instrument. Variations of the base substrate **131** can, however, include other elements. For instance, as described in more detail below, the base substrate can include one or more openings, recesses, and/or protrusions that provide further coupling with the sample processing chip **132**, in order to collectively define valve regions for opening and closing flow through the sample processing chip **132**.

(62) As shown in FIGS. 4A and 4C (bottom view), the sample processing chip **132**, (equivalently referred to herein as a microwell device or a slide) defines a set of wells (e.g. microwells). Each of the set of wells can be configured to capture a single cell and/or one or more particles (e.g., probes, beads, etc.), any suitable reagents, multiple cells, or any other materials. In variations, microwells of the sample processing chip **132** can be configured for co-capture of a single cell with a single functional particle, in order to enable analyses of single cells and/or materials from single cells without contamination across wells. Embodiments, variations, and examples of the sample processing chip **132** are described in one or more of: U.S. application Ser. No. 16/048,104, filed 27 Jul. 2018; U.S. application Ser. No. 16/049,057, filed 30 Jul. 2018; U.S. application Ser. No. 15/720,194, filed 29 Sep. 2017; U.S. application Ser. No. 15/430,833, filed 13 Feb. 2017; U.S. application Ser. No. 15/821,329, filed 22 Nov. 2017; U.S. application Ser. No. 15/782,270, filed 12 Oct. 2017; U.S. application Ser. No. 16/049,240, filed 30 Jul. 2018; U.S. application Ser. No. 15/815,532, filed 16 Nov. 2017; U.S. application Ser. No. 16/115,370, filed 28 Aug. 2018, U.S. application Ser. No. 16/564,375, filed 9 Sep. 2019, and U.S. application Ser. No. 16/816,817, filed 12 Mar. 2020, which are each incorporated in their entirety by reference above.

(63) In material composition, the sample processing chip **132** can be composed of microfabricated silicon or glass-fused silica materials, which function to enable higher resolution of the set of wells, enabled, for instance, by defining sharper edges (e.g., thinner well walls, well walls arranged at an angle approaching 90 degrees, etc.) in the set of wells. Materials and fabrication processes described can further enable one or more smaller characteristic dimensions (e.g., length, width,

overall footprint, etc.) of the microwell cartridge as compared to conventional chip designs. Additionally or alternatively, the substrate include any other suitable material, such as—but not limited to—a polymer, metal, biological material, or any other material or combination of materials. Sample processing chip **132** may be fabricated by various processes such as precision injection molding, precision embossing, microlithographic etching, LIGA based etching, or by other suitable techniques.

(64) In some variations, one or more surfaces of the set of wells (e.g., bottom surface, side surface, bottom and side surfaces, all surfaces, etc.) can be reacted with oligonucleotide molecules for capture of biomarkers from individual cells into individual microwells. The oligonucleotide molecules present on each and individual microwells may be barcoded to allow biomarkers processed in each microwell to be linked back to a particular well and hence a particular single cell. In one variation, the set of wells includes a set of microwells having hexagonal cross sections taken transverse to longitudinal axes of the wells, as described in one or more of the applications incorporated by reference above.

(65) In one variation, as shown in FIG. 4C, the sample processing chip **132** can include an inlet opening **32**, a first fluid distribution network **33** downstream of the inlet opening, for distribution of fluids to a set of microwells **34**, a second fluid distribution network **35** downstream of the set of microwells **34**, and an outlet opening **36** coupled to a terminal portion of the second fluid distribution network **35**, for transfer of waste fluids from the sample processing chip **132**. In this variation, the sample processing chip **132** is coupled to a first side (e.g., under-side) of the base substrate **131** (e.g., by laser welding, glue bonding, solvent bonding, ultrasonic welding or another technique). Coupling of the sample processing chip **132** to the side of the base substrate **131** can enable transfer of heat from the heating and cooling subsystem **150** to the set of microwells **34** and/or other regions of the sample processing chip **132**, where the heating and cooling subsystem **150** is described in more detail below.

(66) The base substrate **131**, as described above, can also include an inlet reservoir **133** (e.g., defined at a second side of the base substrate **131** opposing the first side to which the sample processing chip **132** is coupled). The inlet reservoir functions to receive sample material (e.g., samples containing cells, sample containing barcoded cells, sample containing encapsulated materials, samples containing particles, etc.) and/or sample processing materials from the reagent cartridge **120** described above, for delivery into the inlet opening **32** of the sample processing chip **132**. In variations, the inlet reservoir **133** can be defined as a recessed region within a surface of the base substrate **131**, wherein the recessed region includes an aperture that aligns with and/or seals with the inlet opening **32** of the sample processing chip **132**. The inlet reservoir **133** of the base substrate **131** can interface with upstream fluid containing components and/or bubble mitigating components, as described in one or more of: U.S. application Ser. No. 16/048,104, filed 27 Jul. 2018; U.S. application Ser. No. 16/049,057, filed 30 Jul. 2018; U.S. application Ser. No. 15/720,194, filed 29 Sep. 2017; U.S. application Ser. No. 15/430,833, filed 13 Feb. 2017; U.S. application Ser. No. 15/821,329, filed 22 Nov. 2017; U.S. application Ser. No. 15/782,270, filed 12 Oct. 2017; U.S. application Ser. No. 16/049,240, filed 30 Jul. 2018; U.S. application Ser. No. 15/815,532, filed 16 Nov. 2017; U.S. application Ser. No. 16/115,370, filed 28 Aug. 2018, U.S. application Ser. No. 16/564,375, filed 9 Sep. 2019, and U.S. application Ser. No. 16/816,817, filed 12 Mar. 2020, which are each incorporated in their entirety by reference above.

(67) The inlet reservoir **133** can also be configured to interface with a fluid level detection subsystem **159** supported by or otherwise interfacing with the deck **110**, as described in more detail below. In particular, portions of the inlet reservoir **133** can be composed of materials that enable sensing of fluid levels within the inlet reservoir **133** (e.g., by optical interrogation, by pressure sensing, by weight sensing, etc.). For instance, the inlet reservoir **133** can be composed of an optically transparent or translucent material to visible spectrum electromagnetic radiation and/or non-visible spectrum electromagnetic radiation (e.g., by fabrication with different materials, by

fabrication to produce thin regions of material at the inlet reservoir **133**, etc.), where sensing elements of the fluid level detection subsystem **159** can be configured to interrogate a level of fluid within the inlet reservoir **133** accordingly.

(68) In variations, one or more of the inlet reservoir **133** of the base substrate **131** and the inlet **32** of the sample processing chip **132** can include valve components that can be open or closed by one or more components of the system **100**. In a first variation, the inlet reservoir **132** includes an aperture that can be accessed by a pipette tip or any other suitable attachment of a fluid handling subsystem coupled to the gantry **170** (described in more detail below). In some embodiments, the aperture can be closed and therefore prevent fluid from traveling from the inlet reservoir **132** to the sample processing chip **132**. The inlet reservoir **132** can, however, be configured in another suitable manner. The opening associated with the inlet reservoir **133** may have a conical shape surface open towards the top allowing interfacing and sealing a pipette tip such that fluid (aqueous solutions or oil or air) may be pumped directly into the microchannel defined in **33** in FIG. **4C**.

(69) As shown in FIGS. **4A** and **4B**, the base substrate **131** can also define an access region **134** for accessing one or more regions of the sample processing chip **132**, where the access region can allow regions of the sample processing chip **132** to be observed and/or extracted from the sample processing chip **132** at various phases of sample processing. As shown in FIGS. **4A** and **4B**, the access region **134** can be defined as a recessed region within the base substrate **131**, and include an opening **37** aligned with the region of the sample processing chip **132** that includes the set of microwells. The sample processing chip **132** may have as few as 100 microwells to as many as 100 million microwells. As such, in variations wherein the microwell region is open to the environment (e.g., without a covering to seal the wells), the opening **37** of the access region **134** can function as a microwell to provide access to contents of the microwells for observation and/or material extraction (e.g., by magnetic separation, as described in further detail below). The opening **37** can match a morphology and footprint of the microwell region, and in a first variation, as shown in FIG. **4B**, can be a square opening. However, in other variations, the opening **37** can have another suitable morphology.

(70) As shown in FIGS. **4A-4C**, the base substrate **131** can include or otherwise couple to a lid **135** covering the access region **134**, where the lid **135** can include a gasket **136** providing sealing functions, and where the lid **135** functions to transition the access region **134** between open and closed modes, thereby preventing evaporative sample loss and/or contamination of contents of the sample processing chip **132** during operation. The lid **135** can additionally or alternatively function to protect the contents of the microwells or other processing regions of the sample processing chip **132** from debris, enable a processing of the contents of the sample processing chip **132** (e.g. by isolating regions from the ambient environment), initiate the start of a protocol (e.g., by opening to accept reagents from a pipettor), prevent user manipulation of the sample processing chip **132** (e.g., by closing after all necessary reagents have been added), define (e.g., with the lid **135**) part or all of a fluid pathway, cavity, or reservoir (e.g. serve as the top surface of a fluidic pathway between the inlet and the set of microwells, serve as a boundary of a fluid pathway adjacent the microwell region, serve as the top surface of a fluidic pathway between the set of wells and the waste chamber, etc.), or perform any other suitable function.

(71) As shown in FIG. **4B**, in at least one variation, the lid **135** can be complementary in morphology to features of the access region **134**, such that the lid **135** mates with the access region **134**, while providing a gap with the sample processing chip **132**. Additionally, in variations (shown in FIGS. **4B** and **4C**), the lid **135** can be substantially flush with the base substrate **131** at a top surface when the lid **135** is in the closed position. However, the lid **135** can be morphologically configured in another suitable manner.

(72) In variations, a protrusion **38** of the lid **135** can interface with the opening **37** of the access region **134**, thereby substantially preventing access to the opening **37** when the lid is in the closed position. As shown in FIG. **4B**, in some variations, the protrusion **38** can have a base (or other

region) surrounded by a gasket **136**, which functions to seal the opening **37** of the access region **134** in the closed position of the lid **135**. Variations of the lid **135** can, however, omit a gasket and promote sealing of the access region **134** in another suitable manner.

(73) In some variations, the lid **135** can include a locking or latching mechanism that allows the lid **135** to be maintained in the closed position with the base substrate **131** until the locking/latching mechanism is released. In the variation shown in FIGS. 5A-5C, a peripheral portion of the lid **135** can include a one or more tabs **39** that interface with corresponding tab receiving portions of the base substrate **131**, where, the tabs **39** are configured to flex when pushed into the base substrate **131** until they interface with the tab receiving portions of the base substrate **131** and return from a flexed configuration to a latched state. Additionally or alternatively, in the variation shown in FIGS. 5A-5C, the locking/latching mechanism can include a releasing body **41** (e.g., bar, recess, hook, etc.) that can be interfaced with in order to release the tab(s) **39** from the tab receiving portions, and transition the lid **135** from the closed mode to the open mode in relation to the base substrate **131**. As such, the lid **135** provides the lid an open mode in which the access region **134** is uncovered and a closed mode in which the access region **134** is covered. In the variation shown in FIGS. 5A-5C, the releasing element **41** includes a bar that is recessed away from the access region **134** of the base substrate **131**, where the bar can be reversibly coupled to a lid-opening tool **145**. In variations, the lid-opening tool **145** can include a first region (e.g., first end) that interfaces with an actuator (e.g., actuating tip, pipettor of a fluid handling subsystem coupled to the gantry **170** described below, etc.), and a second region (e.g., second end) including a linking element **42** configured to interface with the releasing element **41** of the lid **135**. Then, with movement of the pipettor/pipette interface, the lid-opening tool **145** can be configured to pull on the releasing element **41** and/or push on the lid **135** in order to transition the lid between open and/or closed modes. As such, in relation to fluid handling elements coupled to the gantry **170** described below, the system Dm can provide operation modes for: coupling a lid-opening tool **145** to an actuator (e.g., coupled to a gantry **170**), the lid-opening tool including a linking element **42**; moving the lid-opening tool into alignment with a releasing element **41** of the lid **135**, reversibly coupling the linking element **42** with the releasing element **41**; and applying a force to the releasing element **41**, thereby releasing the lid **135** from a latched state and transitioning the lid **135** from a closed mode to an open mode. In order to effectively apply an unlatching force (e.g., by the actuator (e.g., coupled to a gantry **170**), the base substrate **131** can be retained in position (e.g., by retention elements described in Section 2.1.4, by retention elements of the heating and cooling subsystems, by retention elements of the fluid level detection subsystem, by retention elements of the deck, etc.) which passively or actively apply counteracting forces against the unlatching forces applied through the lid-opening tool **145**.

(74) In variations, however, the locking/latching mechanism can additionally or alternatively include or operate by way of: a lock-and-key mechanism, magnetic elements, or another suitable mechanism. Furthermore, in alternative variations, the lid **135** can include another lid actuator, for instance, including a motor that rotates the lid about an access parallel to a broad surface of the sample processing cartridge **130**. The actuator can additionally or alternatively be configured to translate the lid **135** (e.g. slide the lid **135** parallel to a broad surface of the sample processing cartridge **130**, translate the lid **135** perpendicular to the broad surface, etc.) or otherwise move the lid **135** to selectively cover and uncover one or more predetermined regions (e.g. the set of microwells). As such, the lid **135** can be configured to operate in an automated or semi-automated fashion, such that the lid **135** automatically closes upon one or more triggers (e.g., cell capture protocol is initiated by a user, cell processing protocol is initiated by a user, all reagents for a selected protocol have been added from the reagent cartridge **120**, etc.) and opens upon one or more triggers (e.g., cell capture protocol has been completed, upon user request, it has been determined that the cells are viable, it has been determined that single cells have been captured, etc.). Additionally or alternatively, operation of the lid **135** can be initiated and/or completed by a

user, operated according to a schedule or other temporal pattern, or otherwise operated.

(75) As shown in FIGS. 4A-4C, the base substrate **131** can also include a waste containment region **137** for receiving waste material from the sample processing chip **132**. The waste containment region **137** can also function to maintain desired pressures (e.g., vacuum pressures, etc.) within the sample processing chip **132**, thereby enabling flow of liquid from the inlet reservoir **133** through the sample processing chip **132** and to the waste containment region **137**. The waste containment region **137** can be defined as a volume (e.g., recessed into the base substrate **131**, extending from the base substrate **132**, coupled to an outlet of the base substrate **131**, etc.) for receiving waste or other materials from the sample processing chip **132**. In the variation shown in FIGS. 4A-4C, the waste containment region **137** is defined at a side of the base substrate **131** opposing the side to which the sample processing chip **132** is coupled, such that waste from the sample processing chip **132** is pushed or pulled upward into the waste containment region **137** by forces of the pumping subsystem **157** described in more detail below. However, the waste containment region **137** can additionally or alternatively be configured in another suitable position relative to the base substrate **131** and the sample processing chip **132**, in order to receive waste.

(76) The waste containment region **137** can have a volumetric capacity of 10-100 mL or another suitable volumetric capacity.

(77) As shown in FIGS. 4A-4C, the waste containment region **137** can include a cover **48** (e.g., a cover that is approximately co-planar with the lid **135**), which facilitates containment of waste within the waste containment region **137**. Alternatively, the waste containment region **137** may not include a cover. Furthermore, as shown in FIG. 4C, examples of the waste containment region **137** can include a pump outlet **51** distinct from the cover, where the pump outlet **51** can allow the residual air in the waste chamber to be pressurized by an off-cartridge pump (e.g., by pumping mechanisms, etc.); however, variations of the waste containment region **137** can alternatively omit a waste outlet.

(78) In relation to the waste containment region **137**, the system **100** can further include a valve **43** configured to allow and/or prevent flow from the sample processing chip **132** to the waste containment region **137**. The valve **43** can interface with the outlet opening **36** of the sample processing chip **132** described above, in order to enable and/or block flow out of the outlet opening **36** and into the waste containment region **137**. The valve **43** can have a normally open state and transition to a closed state upon interacting with a valve-actuating mechanism. Alternatively, the valve **43** can have a normally closed state and transition to an open state upon interacting with a valve-actuating mechanism.

(79) In the variation shown in FIGS. 4A and 6A-6B, the valve **43** comprises an elastomeric body and is configured to couple the sample processing chip **132** to the base substrate **131** through an opening **44** of the sample processing chip **132** that aligns with a corresponding valve-receiving portion of the base substrate **131**. In this variation, a transitionable portion of the valve **43** is configured to be positioned along a flow path from the outlet opening **36** of the sample processing chip **132** to the inlet of the waste containment region **137** of the base substrate **132** (e.g., along a flow path from the microwell region to an outlet of the sample processing chip into a waste containment region of the sample processing cartridge). In an example the opening **44** of the sample processing chip **132** is contiguous with the outlet opening **37** of the sample processing chip **132**; however, in other variations, the outlet opening **37** and the opening **44** may be displaced from each other and connected by another microfluidic channel. As such, closure of the valve **43** can block flow from the outlet opening **37** into the waste containment region **137**, and the valve **43** can be opened to allow flow from the outlet opening **37** into the waste containment region **137**.

(80) In a variation shown in the cross sectional images of the base substrate **131** shown in FIG. 6B, a valve actuator **45** can access the base substrate **131** from below (e.g., from below the deck), and pass through a channel or other recess/opening of the base substrate **132** in order to interact with the valve **43**. In particular, when a tip **46** (aligned with the opening into the base substrate) of the

valve actuator **45** pushes against the valve (e.g., an elastomeric membrane of the valve **43**), as shown in FIG. **6B** (top), the valve **43** can transition to a closed state in order to fluidically decouple the outlet opening **37** of the sample processing chip **132** from the waste containment region **137**. Additionally or alternatively, as shown in FIG. **6B** (bottom), removal of force by the valve actuator **45** can remove pressure from the valve **43** and transition it to an open state to fluidically couple the outlet opening **36** of the sample processing chip **132** from the waste containment region **137**. As such, the valve actuation subsystem includes an engaged mode wherein the tip extends into the valve opening to deform the elastomeric valve, thereby closing the flow path, and a disengaged mode wherein the tip is retracted, thereby opening the flow path. However, the valve **43** can additionally or alternatively be configured in another suitable manner.

(81) In other variations, the system can include a similar mechanism for coupling a valve to other flow paths of the sample processing chip **132** and/or to the base substrate **131**.

(82) Variations of the base substrate **131** can, however, include other elements. For instance, as described in more detail below, the base substrate **131** can include one or more openings, recesses, and/or protrusions that provide further coupling with the sample processing chip **132**, in order to promote or inhibit flow through the sample processing chip **132**. For instance, as shown in FIG. **6A**, the base substrate can include a pump opening **46** that couples the base substrate **131** to a pumping element of the pumping subsystem **157** (e.g., through deck no), in order to drive and/or stop fluid flow through the sample processing chip **132**.

(83) The base substrate **131** of the sample processing cartridge **130** can, however, include other suitable elements.

(84) 2.1.3 Deck-Supported Element: Tool Container

(85) As shown in FIGS. **2A** and **2B**, the deck **110** includes at least one region **113** for supporting a unit of the tool container **140**, where the region **113** functions to position the tool container **140** relative to fluid handling apparatus of the gantry **170** described below. The region **113** can also position the tool container **140** in proximity to the reagent cartridge **120** and sample processing cartridge **130** being used, in order to provide a more compact system and improve efficiency of automated operations involving contents of one or more of the reagent cartridge **120**, sample processing cartridge **130**, and tool container **140**. The region **113** can also include an opening that allows the tool container **140** to be at least partially recessed below a surface of the deck **110**.

(86) The tool container **140** functions to contain, in one or more compartments, one or more units of various tools for fluid aspiration, fluid delivery, separation of target material from non-target material of a sample, sample processing cartridge lid-opening tools **145**, and/or other tools, according to one or more workflows for various applications. As such, the tool container **140** can facilitate transfer and/or mixing of reagents with sample, fluidically couple and/or decouple elements at various regions of the deck **110**, or otherwise interact with one or more components of the system **100**.

(87) In variations, one of which is shown in FIG. **7A** the tool container **140'** can include a set of tips **141** for fluid aspiration and/or fluid delivery (e.g., to and from the reagent cartridge **120**, to and from the sample processing cartridge **130**, etc.). The set of tips **141** can include any or all of: standard pipette tips (e.g. P20 tips, P200 tips, P1000 tips, etc.); additionally or alternatively, the set of tips can include piercing tools (e.g., to gain access to a reagent tube through a seal), blunt tips (e.g. for facilitating fluid flow, for blocking an aperture, for defining a fluidic pathway, etc.), or any other suitable tips. The tips can be configured or otherwise used in any or all of: piercing (e.g. piercing a reagent strip foil), transferring and/or mixing a set of reagents (e.g. a tip for mixing ethanol and priming buffer, a tip for transferring SPRI ethanol, a tip for transferring SPRI supernatant, a tip for transferring SPRI elution buffer, etc.), cell dispensing into a unit of the sample processing cartridge **130**, dispensing of functionalized particles into a unit of the sample processing cartridge, facilitating the performance of a predetermined set of processes (e.g., particle washing, cell lysis, oil dispensing, pre-reverse-transcription wash, other workflows described below etc.), or

can have any other suitable function.

(88) As shown in FIG. 7A, the set of tips **141** can be arranged in an array (e.g., according to type, according to size, etc.) within the tool container **140**, where the internal base of the tool container **140** can include drip-catching sections for catching drips leaving the aspiration/delivery ends of the tips from various fluid handling operations. Furthermore the tool container **140** can include spacing elements for preventing individual tips of the set of tips **141** from contacting each other, thereby preventing cross-contamination. Furthermore, regions of one or more of the tips for interfacing with the pipettor can be conductive (e.g., thermally conductive, electrically conductive, composed of a metal, composed of a conductive polymer, composed of a semiconducting material, etc.).

(89) In a specific example for a 3' processing protocol shown in FIG. 7B, the tool container **140** can include: a first tip **1401** (e.g., with a tip volume of 200 mL, with a tip volume of 200 μ L) for transferring ethanol, a wash buffer, and/or a priming solution; a second tip **1402** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transferring of a cell suspension; a third tip **1403** (e.g., with a tip volume of 200 mL, with a tip volume of 200 μ L) for transferring functionalized particles; a fourth tip **1404** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transferring a wash buffer; a fifth tip **1405** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of a particle binding buffer; a sixth tip (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of DTI and/or a lysis buffer; a seventh tip **1407** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of perfluorinert oil; an eighth tip **1408** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of DTT, particle binding solution, and/or wash solution; a ninth tip **1409** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of DTT and/or a pre-RT reaction wash solution; one or more magnetic sleeves **1410** for magnetic retrieval; an 11.sup.th tip **1411** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of DTT and/or an RT cocktail solution; a 12.sup.th tip **1412** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of an exonuclease treatment; a 13.sup.th tip **1413** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of mineral oil; a 14.sup.th tip **1414** for transfer of an exonuclease treatment; a 15.sup.th tip **1415** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of a sodium hydroxide solution; a 16.sup.th tip **1416** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of a second wash buffer; a 17.sup.th tip **1417** for transfer of a second strand synthesis solution; an 18.sup.th tip **1418** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of a particle solution; a 19.sup.th tip **1419** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of a second wash buffer; a 20.sup.th tip **1420** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of a PCR master mix and/or particle solution; a 21.sup.st tip **1421** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of mineral oil; a 22.sup.nd tip **1422** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of a PCR product; a 23.sup.rd tip **1423** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of AMPure and/or supernatant; a 24.sup.th tip **1424** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of ethanol; a 25.sup.th tip **1425** (e.g., with a tip volume of 1000 mL, with a tip volume of 1000 μ L) for transfer of ethanol; a 26.sup.th tip **1426** (e.g., with a tip volume of 50 mL, with a tip volume of 50 μ L) for transfer of ethanol; and a 27.sup.th tip **1427** (e.g., with a tip volume of 50 mL, with a tip volume of 50 μ L) for transfer of nuclease-free water and/or derived products.

(90) In variations, the tool container **140** can additionally or alternatively include other sample processing tools. In one variation, the tool container **140** can include one or more units of a separation tool tip **142** for magnetic separation of target material from non-target material (described in more detail below). Additionally or alternatively, in variations, the tool container **140** can additionally or alternatively include units of a lid-opening tool **145** for transitioning the lid **135** described above to an open configuration. However, in relation to disposability of the tool

container **140** and/or its contents, the tool container **140** can be configured to contain only disposable elements, and to omit reusable elements (e.g., units of a lid-opening tool **145**, as shown in FIG. 2B).

(91) Furthermore, units of contents of the tool container **140** can additionally be included with one or more of the reagent cartridge **120**, the sample processing cartridge **130**, otherwise arranged at the deck **110**, separate from the deck **110**, separate from the system **100** or otherwise arranged. In other variations, re-useable tools used in the tool container **140** may include other tools that use electrical, electromagnetic, optical or combination of different modalities to interact with the gantry **170** and be moved to specific locations over the reagent cartridge **120** and/or the sample processing cartridge **130** to provide specific energies (e.g., heat, optical signals, electromagnetic waves, etc.) and/or sense specific signals (e.g., optical, thermal, electromagnetic, etc.). These tools may be wired to the control electronics or may be wirelessly charged and controlled/communicated.

(92) 2.1.4 Deck-Supported Elements—Registration and Retention Features

(93) As shown in FIGS. 8A through 8I, the deck no can include a set of retention elements positioned relative to region in for the reagent cartridge **120**, region **112** for the sample processing cartridge **130**, and region **113** for the tool container **140**, where the set of retention elements function to register and retain the reagent cartridge **120**, sample processing cartridge **130**, and tool container **140** at the deck **110** during operation to process samples, and to enable release of the reagent cartridge **120**, sample processing cartridge **130**, and tool container **140** from the deck **110** when appropriate. The set of retention elements can include mechanical retention elements (e.g., recesses, protrusions, etc.) configured provide retention by way of a snap fit, a press fit, ratcheting, or another suitable mechanism. Additionally or alternatively, the set of retention elements can include magnetic retention elements or other suitable retention elements. In variations including magnetic retention elements, associated magnets can be positioned away from magnetic separation areas of the reagent cartridge **120** and deck **110** (described in relation to 2.1.8 below) so as to not interfere with associated magnetic separation mechanisms. Alternatively, the associated magnets can be positioned in proximity to the magnetic separation areas of the reagent cartridge **120** and deck **110**, in order to facilitate magnetic separation operations performed at the reagent cartridge **120** (and/or other suitable portions of the system).

(94) The retention elements can provide uniform mechanisms for each of the reagent cartridge **120**, sample processing cartridge **130**, and tool container **140**. Alternatively, the reagent cartridge **120**, sample processing cartridge **130**, and tool container **140** can each include different retention elements that operate by different mechanisms as appropriate. The retention mechanisms supported by the set of retention elements can be manually operated (e.g., a user interacts with the retention elements to disengage and/or engage a component with the deck no). Additionally or alternatively, the retention mechanisms supported by the set of retention elements can be non-manually operated (e.g., with actuators coupled to the retention elements in order to transition them between engaged and disengaged modes). Retention mechanisms are configured to operate with appropriate morphologies (e.g., to facilitate engagement by a manual operator or apparatus), loading and unloading forces, and/or transmitted forces (e.g., to other sensitive elements of the deck **110**).

(95) In the variation shown in FIG. 8A, the deck **110** can include a first subset of retention elements **21** corresponding to region in for the reagent cartridge **120**, a second subset of retention elements **22** corresponding to region **112** for the sample processing cartridge **130**, and a third subset of retention elements **23** corresponding to region **113** for the tool container **140**.

(96) In relation to FIG. 8A, the first subset of retention elements **21** at region **111** can include a set of snap ledges positioned about region **111** (e.g., at contralateral sides of region in), in a manner that complements corresponding retention interfaces **31** of the reagent cartridge **120**. In more detail, as shown in FIGS. 8B and 8C, the retention interfaces **31** of the reagent cartridge **120'** can include a set of tabs (e.g., flexible tabs) that can be compressed from a baseline position to be positioned at region in, where release from the compressed configuration allows the set of tabs to latch with the

snap ledges. The set of tabs of the retention interfaces **31** can further include gripping features (e.g., bumps, etc.) that allow a user to compress the set of tabs easily. In related variations, the first subset of retention elements **21** and corresponding retention interfaces of the reagent cartridge **120'** can be positioned in another manner. For instance, the set of tabs of the reagent cartridge can be positioned at a non-peripheral region of the reagent cartridge **120** (e.g., with finger and/or thumb hole cutouts), with corresponding alignment with the retention elements **21** of the deck **110**. In relation to providing robust contact, the retention elements **21** can be configured to provide a biasing force against deck elements (e.g., heating and cooling subsystem elements, magnetic separation subsystem elements, etc.) that the reagent cartridge **120** interacts with, in order to provide robust contact between surfaces. The biasing force used against the deck elements may be at least 0.5 lbs or at least 1 lbs or at least 2 lbs, or at least 3 lbs.

(97) In relation to FIG. **8A**, the second subset of retention elements **22** at region **112** can include a set of snap ledges positioned about region **112** (e.g., at contralateral sides of region **112** and at a third peripheral side of region **112**), in a manner that complements corresponding retention interfaces **32** of the sample processing cartridge **130**. In more detail, as shown in FIGS. **8D** through **8G**, the retention interfaces **32** of the sample processing cartridge **130** can include a set of tabs (e.g., flexible tabs) that can be compressed from a baseline position to be positioned at region **112**, where release from the compressed configuration allows the set of tabs to latch with the snap ledges. In this variation, the sample processing cartridge **130** can be configured to be inserted into/engaged with a first of the retention elements **22a** (e.g., at a side of the sample processing cartridge near the inlet reservoir, at a side of the sample processing cartridge near the waste containment, region, etc.) prior to latching of the other retention interfaces **22b** with the contralateral snap ledges. Similar to the set of tabs of the reagent cartridge **120**, the set of tabs of the retention interfaces **32** of the sample processing cartridge **130** can further include gripping features (e.g., bumps, etc.) that allow a user to compress the set of tabs easily. In relation to providing robust contact, the retention elements **22** can be configured to provide a biasing force against deck elements (e.g., heating and cooling subsystem elements, pumping subsystem elements, etc.) that the sample processing cartridge **130** interacts with, in order to provide robust contact between surfaces. Furthermore, the set of retention elements **22** of the deck no can be configured to reversibly engage the sample processing cartridge **120** and provide a counteracting force against the lid-opening tool **145** described in relation to lid-opening operation modes.

(98) In relation to FIG. **8A**, the third subset of retention elements **23** at region **113** can include a set of snap ledges positioned about region **113** (e.g., at contralateral sides of region **113**), in a manner that complements corresponding retention interfaces **33** of the tool container. In more detail, as shown in FIGS. **8H** and **8I**, the retention interfaces **33** of the tool container **140** can include a set of tabs (e.g., flexible tabs) that can be compressed from a baseline position to be positioned at region **113**, where release from the compressed configuration allows the set of tabs to latch with the snap ledges. The set of tabs of the retention interfaces **33** can further include gripping features (e.g., bumps, etc.) that allow a user to compress the set of tabs easily.

(99) In variations of the retention elements of the deck no described above can, however, include other suitable features or be configured relative to associated elements in another suitable manner.

(100) 2.1.5 Deck-Supported Element: Heating and Cooling Subsystem

(101) As shown in FIGS. **1A-1D** and **2B**, the system **100** can include a heating and cooling subsystem **150**, which functions to transfer heat to and/or from desired regions of the reagent cartridge **120** and/or the sample processing cartridge **130**. The heating and cooling subsystem **150** can additionally or alternatively function to maintain desired temperatures within internal volumes of the system **100**. In variations, the heating and cooling subsystem **150** can include one or more units of: heating elements (e.g., Peltier heating elements, resistive heating elements, other heating elements), cooling elements (e.g., Peltier cooling elements, chilled aluminum block, fluidic pathway system to circulate coolant, etc.), thermal contact or non-contact bodies for transferring

heat to or from the heating and cooling elements to other objects, heat sinks, fans, temperature sensors, and thermal control circuitry (e.g., with electrical coupling to processing elements of the base **180** described in more detail below). In variations, the cooling element(s) can maintain storage volumes and/or samples between 2 and 8 degrees Celsius, further preferably at 4 degrees Celsius. Additionally or alternatively, the cooling elements can maintain one or more storage volumes/samples at any suitable temperature (e.g. below 2 degrees Celsius, above 8 degrees Celsius, etc.).

(102) One or more portions of the heating and cooling subsystem **150** can pass into openings of the deck no to thermally interface with or otherwise couple with desired portions of other system elements (e.g., reagent cartridges, sample processing cartridges, tool container, etc.) supported by the deck **110**, in order to provide heat transfer functions for various applications. Alternatively, the deck no can be composed of a thermally conductive material at desired regions for heat transfer applications, and portions of the heating and cooling subsystem **150** can be configured to contact the thermally conductive material regions of the deck **110** for heat transfer.

(103) In the specific example shown in FIG. 2B, the heating and cooling subsystem **150** includes a set of thermal bodies that thermally interface the heating elements with desired portions of the reagent cartridge **120** and the sample processing cartridge **130** supported by the deck **110**. The set of thermal bodies includes a first thermal body **156'** (e.g., thermal plate) passing through an opening of the deck **110'** and configured to interface with the first domain **121'** of the reagent cartridge **130'** described above, where the first thermal body **156'** includes an array of recesses for surrounding storage volumes of the first domain **121'** (e.g., to provide a chilled environment for materials stored in the storage volumes of the first domain **121'**). As shown in FIG. 2B, the set of thermal bodies can also include a second thermal body **157'** (e.g., thermal plate) passing through an opening of the deck **110'** and configured to interface with the third domain **123'** of the reagent cartridge **130'** described above, where the second thermal body **157'** includes an array of recesses for surrounding storage volumes of the third domain **123'** (e.g., to transfer heat for PCR and/or other heating operations for processes conducted within storage volumes of the third domain **123'**). As shown in FIGS. 2B and 6A, the set of thermal bodies can also include a third thermal body **158'** (e.g., thermal plate) passing through an opening of the deck **110'** and configured to interface with the set of microwells **34** of the sample processing chip **132** described above, where the third thermal body **158'** provides a substantially planar substrate for uniform heat transfer to and from microwells of the sample processing cartridge **130**.

(104) In variations, the set of thermal bodies can be coupled to heat sink elements **155** (e.g., at sides of the thermal bodies away from interfaces with reagent cartridges/sample processing cartridges), in order to provide greater surface area for heat transfer. Furthermore, as shown in FIGS. 2D-2F, areas between the deck **110** and the base **180** described in more detail below can include one or more fans **154** and/or ducts **153**, in order to provide thermal mechanisms for convective heat transfer away from the set of thermal bodies as needed. Furthermore, in variations described above, one or more portions of the heating and cooling subsystem **150** (e.g., thermal bodies, etc.) can include features that facilitate retention of corresponding cartridges (e.g., reagent cartridges, sample processing cartridges, etc.) in position.

(105) In variations, one or more of the thermal bodies and/or other portions of the heating and cooling subsystem **150** can be coupled to actuators that move the thermal bodies into and out of thermal communication with elements supported by the deck **110**; however, variations of the system **100** can omit actuators of the heating and cooling subsystem **150**.

(106) 2.1.6 Deck-Supported Element: Pumping Subsystem

(107) As shown in FIGS. 1A, 2B, and 6A, the system **100** can include a pumping subsystem **157** (e.g., coupled to the deck **110** and/or base **180**), which functions to provide positive pressure and/or negative pressure to desired portions of the sample processing cartridge **130** described above. In more detail, the pumping subsystem **157** can function to drive fluid flow from the inlet reservoir

133 and into the sample processing chip **132** of the sample processing cartridge **130**. Additionally or alternatively, the pumping subsystem **157** can function to remove fluid from the waste containment region **137** of the sample processing cartridge **130** and into an external waste receptacle. In variations, the pumping subsystem **157** can include one or more ports **58** (e.g., vacuum ports) configured to interface with the sample processing cartridge **130** through openings in the deck **110**, one or more pumps (e.g., vacuum pumps, peristaltic pumps, etc.) coupled to the ports **58**, one or more manifolds to provide pressure driving pathways coupled to the pump(s), one or more pressure sensors configured to detect pressure levels along pressure pathways, and/or one or more control circuit elements configured to control operation of the pumping subsystem **157** (e.g., with electrical coupling to processing elements of the base **180** described in more detail below). As such, in variations, portions of the pumping subsystem **157** not directly coupled to the sample processing cartridge **130**. can be situated between the deck **157** and the base.

(108) In variations, the port(s) **58** of the pumping subsystem **157** can be coupled (e.g., physically connected, fluidically connected, etc.) suitable regions of the sample processing cartridge **130** (e.g. inlet, wells, etc.), and can additionally or alternatively be coupled to the reagent cartridge **120**, another fluidic pathway of the system **100**, or any other suitable component of the system **100**.

(109) In a first specific example, as shown in FIGS. 2F and 6A, the pumping subsystem **157** includes a pump **59** (e.g., peristaltic pump) arranged between the deck **110** and the base **180**, and coupled to the waste containment region at the waste outlet **51**, through a coupling arranged at a bottom broad surface of the base substrate **131** of the sample processing cartridge **130**, wherein the pump **59** can draw fluid through the sample processing chip **132** in forward and reverse directions by applying negative and/or positive pressure accordingly. Additionally or alternatively, the pumping subsystem **157** can draw fluid from the inlet reservoir **133** and through the sample processing chip **132** at a predetermined pressure (e.g., -0.25 psi or -1 psi or -2.5 psi) according to triggering events associated with the fluid level detection subsystem **159** described in more detail below.

(110) In relation to the sample processing cartridge **130** and forces applied by the lid-opening tool **145** to transition the lid **135** between closed and open states, retention elements described in Section 2.1.4 above can provide a retention force that balances/counteracts a force applied by the lid-opening tool **145** to open the lid **135**, as described above. Additionally or alternatively, one or more portions of the pumping subsystem **157** can retain the sample processing cartridge **130** in position and provide a retention force that balances/counteracts a force applied by the lid-opening tool **145** to open the lid **135**, as described above. In a specific example, the port **58** can couple with the sample processing cartridge **130** to provide a counteracting force to the lid-opening tool **145**. However, other portions of the system **110** can additionally or alternatively provide a counteracting force.

(111) 2.1.7 Deck-Supported Element: Fluid Level Detection Subsystem

(112) As shown in FIGS. 1A, 2B, and 9, the system **100** can include a fluid level detection subsystem **159** at least partially supported by the deck **110** and configured to interface with the sample processing cartridge **130**. The fluid level detection subsystem **159** functions to detect and/or measure a fluid parameter (e.g. a binary presence of fluid, a volume of fluid, a fluid flow rate, a fluid type etc.) associated with fluid at the sample processing cartridge **130** and/or other fluid processing elements of the system **100**. In variations, the fluid level detection subsystem **159** can include a fluid level sensor **63** coupled to fluid level control circuitry (e.g., with electrical coupling to processing elements of the base **180** described in more detail below).

(113) A unit of the fluid level sensor **63** can determine a fluid parameter associated with the inlet reservoir **133** (e.g. fluid passing from the inlet reservoir to microwells) of the sample processing cartridge **130**. A unit of the fluid level sensor **63** can additionally or alternatively determine a fluid parameter associated with an outlet (e.g. fluid passing from microwells to outlet, fluid passing from outlet to waste containment region, etc.) of the sample processing cartridge **130**. A unit of the fluid

level sensor **63** can additionally or alternatively determine a fluid parameter associated with the waste containment region (e.g. volume of fluid in waste containment region) of the sample processing cartridge **130**, or any other suitable fluid parameter. The fluid level sensor **63** can include any or all of: an optical sensor, pressure sensor, temperature sensor (e.g. to detect a fluid of certain temperature in a fluidic pathway), or any other suitable sensor configured to detect the presence of fluid and optionally determine a value associated with the fluid (e.g. volumetric flow rate, etc.) in the sample processing cartridge.

(114) In a first variation, the system includes an optical sensor **63** configured to detect the presence of fluid being transferred from the inlet to the set of wells. In an example, an infrared (IR) emitter/detector pair is used to determine the presence of fluid and a volume of the fluid being transferred (e.g. further based on the duration of time that the fluid is present) at the inlet reservoir **133** of the sample processing cartridge **130**.

(115) In the specific example shown in FIGS. **2B** and **9**, the fluid level detection subsystem **159** includes an IR emitter **63a** displaced from and opposing an IR detector **63b** across the inlet reservoir **133** of the sample processing cartridge **130**, where the IR emitter **63a** and IR detector **63b** are configured to be positioned within an underside internal portion of the base substrate **131** of the sample processing cartridge **130**, to pass IR radiation across walls and an internal volume of the inlet reservoir **133**, thereby enabling fluid level detection. Variations of the fluid level sensor **63** can, however, be configured in another suitable manner.

(116) 2.1.8 Deck-Supported Element: Separation Subsystem and Operation Modes

(117) As shown in FIGS. **1A**, **2B**, and **10A-10C**, the system **100** can include a separation subsystem **160**, which functions to facilitate separation of target material from non-target material (e.g., using magnetic forces, using other forces). In variations, the separation subsystem **160** can include embodiments, variations, and examples of components described in U.S. Application 62/866,726, titled "System and Method for Target Material Retrieval from Microwells" and filed on 26 Jun. 2019, which is herein incorporated in its entirety by this reference. However, variations of the separation subsystem **160** can additionally or alternatively include other components.

(118) In one variation, as shown in FIGS. **10A-10C**, the separation subsystem **160** can include a first body **161** including an interface **162** to the fluid handling subsystem (e.g., pipette interface) of the gantry **170** described below, and a magnetic distal region **163** configured to provide magnetic forces for target material separation. In this variation, the magnetic distal region **163** can be configured to couple with one or more units of the magnetic sleeves **1410** (e.g., of tool container **140**) described above, where the magnetic sleeves **1410** can be disposable elements. Furthermore, the interface **162** can be configured to couple to a pipetting head coupled to the gantry **170** described in more detail below, in order to facilitate target or non-target material retrieval by way of magnetic forces, fluid aspiration, and/or fluid delivery operations provided by the pipetting head. As such, the system **100** can include a separation mode in which the gantry **170** transports the first body **161**, coupled to the magnetic sleeve **1410**, between the sample processing cartridge **130** and the reagent cartridge **120** for magnetic separation of target material from a sample. Furthermore, embodiments of methods implemented using the separation subsystem **160** can produce rapid retrieval of target material, with a retrieval efficiency of >90% where only magnetic particles coupled to target material (or non-target material) of the sample are retrieved. The separation subsystem **160** can thus function to produce increased selective retrieval efficiency can thus reduce downstream costs in relation to processing reagent and other material costs (due to reduced volumes needed, due to reduced splits in biochemistry reactions) and processing burden.

(119) As shown in FIG. **10A**, the first body **161** can include an interface **162** to the fluid handling subsystem of the gantry **170** described below, where the interface includes a coupling region that complements a corresponding coupling region of the fluid handling subsystem. The coupling region of the interface **162** can operate by: a magnetic coupling mechanism; a press fit; a snap fit, a screwing mechanism; a male-female connection; or another suitable mechanism for providing

reversible coupling with the fluid handling subsystem.

(120) The magnetic distal region **163** of the first body **161** can include or be composed of a material for providing a permanent magnet, or can alternatively be configured as an electromagnet (e.g., with coupling to suitable electronics of the system **Dm**). In variations, the magnetic distal region **163** can be composed of one or more of: alnico, neodymium, neodymium iron boron, samarium cobalt, ferrite, and any other suitable magnetic material. In morphology, the magnetic distal portion **163** can complement a morphology of the magnetic sleeve **1410**, such that units of the magnetic sleeve **1410** can couple (e.g., reversibly couple) with the magnetic distal portion. Furthermore, the morphology and pole configuration of the magnetic distal portion **163** is such that nearly normal magnetic force is applied to majority of the target microwells from where entrapped particles are being removed.

(121) The magnetic sleeve **1410** can include a first region **1410a** configured to interface with the sample processing chip **132**, for instance, through access region **134**, in order to enable transfer of material from the sample processing chip **132**. The magnetic sleeve **1410** can also include a second region **1410b** for coupling with the magnetic distal portion **163** of the first body **161**, and an internal cavity **1410c** passing from the first region to the second region. The magnetic sleeve **1410** functions to provide structures that separate the first body **163** from physically contacting wells or other sensitive material of the sample processing chip **132**, and to support application of a magnetic field to the desired regions for retrieval of target material (or non-target material). The magnetic sleeve **1410** can also function to prevent sample cross contamination, by serving as a disposable component that can be discarded between uses of the system **100**.

(122) The magnetic sleeve **1410** can be morphologically prismatic with an internal cavity **1410c**, where the cross section of the magnetic sleeve **1410** along its longitudinal axis is defined by a polygonal perimeter, an ellipsoidal perimeter, an amorphous perimeter, or a boundary of any other suitable shape (e.g., closed shape, open shape). The cross section of the magnetic sleeve **1410** can complement a shape of a footprint of the microwell region of the sample processing chip **132**, but may alternatively not complement a shape corresponding to the sample processing chip **132**. The magnetic sleeve **1410** preferably has a wall thickness that supports application of a magnetic force, from the magnetic distal portion **163**, to the sample processing chip **132** interfacing with the first region **1410a** of the magnetic sleeve **1410**. The wall thickness can be constant or non-constant along the length of the adaptor **210**. In examples, the wall thickness can range from 0.2 to 3 mm thick; however, in other examples, the wall thickness can have any other suitable thickness. The surface of the magnetic sleeve **1410** that receives the magnetic particles is made smooth (say surface finish better than SPIB1) such that the small magnetic particles (1-3 micron) do not get entrapped in the surface during the bead capture onto its surface and subsequent release to another receptacle.

(123) The magnetic sleeve **1410** can additionally or alternatively include structural features that enable separation operation modes of the separation subsystem **160**. For instance, in relation to release of the magnetic sleeve **1410** from the pipetting head (described in more detail below), the magnetic sleeve **1410** can include a protrusion **1410d** configured to allow another object (e.g., sleeve stripping tool **165**) to provide a force against the protrusion **1410d** to release the magnetic sleeve **1410** from the first body **161**.

(124) As described above, the magnetic sleeve **1410** couples, at a first region **1410a**, to a region of the sample processing chip **132** exposed through access region **134**, in order to facilitate application of magnetic force to the region, and to enable drawing of material (e.g., target or non-target material coupled to magnetic particles) into the magnetic sleeve **1410** for further downstream processing. The magnetic sleeve **1410** can thus include a seal at the first region **1410a**, in order to facilitate mechanisms for drawing target material from the sample processing chip **132** into the magnetic sleeve **1410**. The seal can be a separate element or an element integrated with the magnetic sleeve **1410**. The magnetic sleeve **1410** can, however, omit a seal at the first region

1410a.

(125) The magnetic sleeve **1410** can be composed of a polymeric material (e.g., plastic) that does not adversely affect the magnetic field applied by the magnetic distal portion **163** during operation. The magnetic sleeve **1410** can additionally or alternatively include (e.g., include particles of) or be composed of a material (e.g., metallic material) that is magnetic or can produce an induced magnetic field to support applications of use of the system **100**. The magnetic sleeve **1410** can additionally or alternatively be composed of any other suitable material. Distributions of the material(s) of the magnetic sleeve **1410** can be homogenous or non-homogenous through the body of the adaptor, in relation to desired magnetic effects at the capture region of the sample processing chip **132**. The internal cavity of the magnetic sleeve **1410** can include a medium (e.g., magnetic medium, etc.), or can alternatively not include any medium.

(126) As shown in FIGS. **1A**, **2B**, and **11A-11B**, in variations, the separation subsystem **160** can include a magnet subsystem **166** including a set of magnets **167** within a housing **168**, where the magnet subsystem **166** further includes a magnet actuator **169** configured to move the set of magnets **167** relative to the deck **110** (e.g., through an opening in the deck), and into/out of alignment with one or more separation reservoirs **129** of the reagent cartridge **120** described above. The magnet actuator **169** can also be coupled to control circuitry (e.g., at the base **180**) described in more detail below. Furthermore, the magnet actuator **169** can be configured to transition the set of magnets between a retracted state and an extended state, wherein in the extended state, the set of magnets passes into the first region of the deck (e.g., as shown in FIGS. **11A** and **11B**). As such, the separation subsystem **160** can also include elements that are supported by the deck **110** and/or base **180**, in order to enable operations for separating target material from non-target material.

(127) In variations, the set of magnets **167** can include one or more permanent magnets and/or electromagnets (e.g., with coupling to suitable electronics of the system **100**). Permanent magnets can be composed of one or more of: alnico, neodymium, neodymium iron boron, samarium cobalt, ferrite, and any other suitable magnetic material.

(128) In the example shown in FIGS. **11A-11B**, the set of magnets **167** can include a first subset of magnets **167a** arranged in a linear array (e.g., for performance of purification operations at the reagent cartridge **120**), where the positions of the first subset of magnets **167a** correspond to positions of volumes of the fifth domain **125'** for particle separation/purification, described in relation to the reagent cartridge **120'** above and workflows in Section 3 below. In the example shown in FIG. **11A-11B**, the set of magnets **167** also includes a second subset of magnets **167b** (e.g., one or more magnets) displaced from or otherwise offset from an axis associated with the first subset of magnets **167a**, in order to interact with a separation reservoir **129a** of the reagent cartridge **120** (e.g., for initial bead retrieval). The set of magnets **167** can, however, be arranged in another suitable manner (e.g., in relation to distributed arrays, in relation to number, etc.) in relation to providing suitable interactions with separation reservoirs **129** of the reagent cartridge **120** or other cartridges.

(129) The housing **168** functions to surround the set of magnets **167**, and to provide smooth operation in relation to transitioning the set of magnets **167** into/out of alignment with corresponding portions of the reagent cartridge **120**. Thus, as shown in FIG. **11B**, in relation to configurations where there is a first subset of magnets **167a** and a second subset of magnets **167b**, the housing **168** can include a first surface (e.g., first planar surface) tracking the first subset of magnets **167a**, and a second surface (e.g., second planar surface) tracking the second subset of magnets **167b**, wherein the first surface **168a** and the second surface **168b** are angled away from each other. In this variation, a pair of opposing walls can extend from the first surface and the second surface, in order to promote smooth operation (e.g., sliding operations) of the housing **168** and magnets through the deck **110** in order to interface with the reagent cartridge **120**.

(130) In relation to the reagent cartridge **120'**, as shown in FIGS. **3A** and **11B**, volumes of the reagent cartridge **120** configured for magnetic separation can each include a planar surface **128a**, or

other surface complementary to the housing **168** at sides configured to be closest to the housing **168** during operation (e.g., in the extended magnet states). Furthermore, volumes of the reagent cartridge **120** configured for magnetic separation can each include a second surface **128b** (e.g., curved surfaces) displaced away from the housing **168** for aspiration and/or delivery of fluids by a pipettor coupled to the gantry **170** described below. Cross sections taken longitudinally through separation volumes/reservoirs **129** of the reagent cartridge **120** can further be tapered toward a base of the reagent cartridge **120**, such that separation operations require a lower volume of fluid and/or provide more efficient aspiration and separation of target from non-target material.

(131) As shown in FIGS. **12A** through **12J**, the separation subsystem **160** can provide a sequence of operation modes for material separation, where, as shown in FIG. **12A**, the operation modes involved specific system structure configurations of: a first body **161** coupled with a pipetting head or other actuatable component (e.g., described in relation to the gantry **170** below), the first body having a magnetic distal region **163**, a magnetic sleeve **1410**, a sleeve stripping tool **165**, a separation reservoir **129**, and a magnet **167b** of the set of magnets **167** described above.

(132) In more detail, as shown in FIG. **12B**, the separation subsystem **160** can provide a first operation mode **164a**, where the first operation mode **164a** is a baseline operation mode in which the first body **161** is uncoupled from a pipette interface or other actuatable component (e.g., described in relation to the gantry **170** below) and the magnetic distal region **163** of the first body **161** is uncoupled from the magnetic sleeve **1410**. The magnetic sleeve **1410** is further retained by sleeve stripping tool **165** above the separation reservoir **129** (or in variations, at another position), and the magnet **167b** is displaced away from the separation reservoir **129** (e.g., by magnet actuator **169** described above).

(133) As shown in FIG. **12C**, the separation subsystem **160** can provide a second operation mode **164b**, where the second operation mode **164b** is an initializing operation mode in which the first body **161** is coupled with a pipette interface or other actuator interface **6** (e.g., described in relation to the gantry **170** below) and the magnetic distal region **163** of the first body **161** is uncoupled from the magnetic sleeve **1410**. The magnetic sleeve **1410** is further retained by sleeve stripping tool **165** above the separation reservoir **129**, and the magnet **167b** is displaced away from the separation reservoir **129** (e.g., by magnet actuator **169** described above).

(134) As shown in FIG. **12D**, the separation subsystem **160** can provide a third operation mode **164c**, wherein, in the third operation mode **164c**, the first body **161** is coupled with a pipette interface or other actuator interface **6** (e.g., described in relation to the gantry **170** below) and moved into alignment with the separation reservoir **129**. In the third operation mode **164c**, the magnetic distal region **163** of the first body **161** is coupled with the magnetic sleeve **1410** above the separation reservoir **129** in the retained position of the magnetic sleeve **1410**. In the third operation mode **164c**, magnet **167b** is displaced away from the separation reservoir **129** (e.g., by magnet actuator **169** described above).

(135) As shown in FIG. **12E**, the separation subsystem **160** can provide a fourth operation mode **164d**, wherein, in the fourth operation mode **164d**, the first body **161** is coupled with a pipette interface or actuator interface **6** (e.g., described in relation to the gantry **170** below) and the magnetic distal region **163** of the first body **161** is coupled with the magnetic sleeve **1410** above the separation reservoir **129**. In the fourth operation mode **164d**, the pipetting head (or other actuatable component) moves the first body **161** and magnetic sleeve **1410** out of the retained position provided by the sleeve stripping tool **165**, to prepare for extraction of material (e.g., functionalized particles from the sample processing cartridge) and/or delivery of material from the pipette interface, coupled to the first body and magnetic sleeve, into the separation reservoir **129**. In the fourth operation mode **164d**, magnet **167b** is displaced away from the separation reservoir **129** (e.g., by magnet actuator **169** described above).

(136) As shown in FIG. **12F**, the separation subsystem **160** can provide a fifth operation mode **164e**, wherein, in the fifth operation mode **164e**, the first body **161** is coupled with a pipette

interface or actuator interface **6** (e.g., described in relation to the gantry **170** below) and the magnetic distal region **163** of the first body **161** is coupled with the magnetic sleeve **1410** above the separation reservoir **129**. In the fifth operation mode **164d**, the pipette interface (or other actuatable component) delivers fluid from the pipetting head into the separation reservoir **129**, and the magnetic sleeve **1410**, still coupled with the first body **161** (and coupled to functionalized particles), is submerged within the fluid in the separation reservoir **129**. In the fifth operation mode **164e**, magnet **167b** is displaced away from the separation reservoir **129** (e.g., by magnet actuator **169** described above).

(137) As shown in FIG. **12G**, the separation subsystem **160** can provide a sixth operation mode **164f**, wherein, in the sixth operation mode **164f**, the first body **161** is coupled with a pipette interface or actuator interface **6** (e.g., described in relation to the gantry **170** below) and the magnetic distal region **163** of the first body **161** is coupled with the magnetic sleeve **1410** above the separation reservoir **129**. In the sixth operation mode **164f**, the pipette interface (or other actuatable component) moves the magnetic sleeve **1410**, still coupled with the first body **161** back into a retained position at the sleeve stripping tool **165**, and the magnetic sleeve **1410** (still coupled with functionalized particles) is submerged within the fluid in the separation reservoir **129**. In the sixth operation mode **164f**, magnet **167b** is displaced away from the separation reservoir **129** (e.g., by magnet actuator **169** described above).

(138) As shown in FIG. **12H**, the separation subsystem **160** can provide a seventh operation mode **164g**, wherein, in the seventh operation mode **164g**, the first body **161** is coupled with a pipette interface or actuator interface **6** (e.g., described in relation to the gantry **170** below) and the magnetic distal region **163** of the first body **161** is coupled with the magnetic sleeve **1410** above the separation reservoir **129**. In the seventh operation mode **164g**, the magnetic sleeve **1410**, still coupled with the first body **161** is in a retained position at the sleeve stripping tool **165**, and the magnetic sleeve **1410** (with functionalized particles) is submerged within the fluid in the separation reservoir **129**. In the seventh operation mode **164g**, magnet **167b** is displaced toward the separation reservoir **129** (e.g., by magnet actuator **169** described above) for to prepare for attraction and retention of target or non-target material coupled to the functionalized particles of the fluid against a wall **128a** of the separation reservoir **129**.

(139) As shown in FIG. **12I**, the separation subsystem **160** can provide an eighth operation mode **164h**, wherein, in the eighth operation mode **164h**, the first body **161** is coupled with a pipette interface or actuator interface **6** (e.g., described in relation to the gantry **170** below), and is being moved away from the separation reservoir **129** to be removed and replaced with a suitable tip from the tool container described above. In the eighth operation mode **164h**, the magnetic distal region **163** of the first body **161** is uncoupled from the magnetic sleeve **1410** above the separation reservoir **129**, by having the pipetting head move the first body **161** away from the magnetic sleeve **1410** while the magnetic sleeve **1410** is retained in position at the sleeve stripping tool **165**. In the eighth operation mode **164h**, the magnetic sleeve **1410** is submerged within the fluid in the separation reservoir **129**. In the eighth operation mode **164h**, magnet **167b** is still positioned in proximity to the separation reservoir **129** (e.g., by magnet actuator **169** described above) for retention of target or non-target material coupled to functionalized particles of the fluid against a wall **128a** of the separation reservoir **129**. In the eighth operation mode **164h**, the magnet **167b** draws target material toward the bottom of the separation reservoir **129** (e.g., for later extraction from the bottom by the pipettor, for retention while the pipettor draws material unbound by the magnet **167b**).

(140) As shown in FIG. **12J**, the separation subsystem **160** can provide a ninth operation mode **164i**, wherein, in the ninth operation mode **164i**, the pipetting head/actuator interface **6** is coupled with a suitable tip and moved into the separation reservoir **129** to aspirate material from the separation reservoir **129**. In the ninth operation mode **164i**, the magnetic sleeve **1410** is still retained in position above the separation reservoir **129** at the sleeve stripping tool **165** and

submerged within the fluid in the separation reservoir **129**. In the ninth operation mode **164i**, magnet **167b** is moved away from the separation reservoir **129** (e.g., by magnet actuator **169** described above) in coordination with aspiration of material by the pipetting head.

(141) FIGS. **13A** through **13D** depict additional views of configurations of a magnetic sleeve **1410** with respect to sleeve stripping tool **165** of a separation reservoir **129** of a reagent cartridge **120**, in relation to operation modes described above.

(142) Variations of the separation subsystem **160** can, however, include elements and provide modes of operation for target material retrieval based upon one or more of: gravitational forces, buoyant forces, centrifugal forces, chemical separation, and/or any other suitable separation approaches. In yet another embodiment, target material retrieval operation by the separation subsystem **160** may be used to transfer target particles from the microwell chip to another substrate or another new empty microwell chip while keeping the relative spatial locations of the different particles being transferred.

(143) 2.2 System—Gantry

(144) As shown in FIGS. **1A-1B**, **2A**, and **2C-2F**, the system **100** can include a gantry **170** coupled to the deck no, which functions to support and/or enable actuation of one or more tools for various interactions with elements of the deck no, along a set of axes. In variations, the gantry **170** provides one or more rails/tracks for moving tools, such as pipettor **174** with pipette interface described below, in three dimensional space (e.g., a three dimensional volume bound by the first side of the deck). In variations, tools actuated using the gantry **170** can be moved relative to the sample processing cartridge **130**, reagent cartridge **120**, tool container **140**, or other elements, for transfer of materials (e.g. cells, reagents, particles, etc.) across different components supported by the deck no. Additionally or alternatively, tools supported by the gantry **170** can be used for reading of barcodes associated with various disposables supported by the deck no (e.g., in relation to identifying proper setup of a run, in relation to inventory management, etc.). The gantry **170** preferably enables movement of one or more tools along one or more axes (e.g., X and Y axes shown in FIG. **2A**) parallel to broad surfaces of the reagent cartridge **120**, sample processing cartridge **130**, and tool container **140**, and additionally along an axis (e.g., Z axis shown in FIG. **2A**) perpendicular to the broad surfaces. The gantry **170** can additionally or alternatively enable movement along a subset of these directions or along any other suitable direction. In order to enable movement, the gantry **170** includes or is otherwise coupled to one or more motors (e.g., motors for each axis or direction of movement), one or more encoders for position identification in each axis or direction of movement, and/or one or more switches (e.g., optical switches for each axis) for control of the gantry **170** (e.g., where the switches are electrically coupled with control circuitry described in relation to the base **180** below).

(145) In a first variation (e.g., as shown in FIGS. **2A-2B** and **2D-2F**), the gantry **170** includes a three dimensional track assembly (X-Y-Z track assembly) including an X-rail **171** for movement of tools along an X-axis, a Y rail **172** for movement of tools along a Y-axis, and a Z-rail **173** for movement of tools along a Z-axis. In the variation shown in FIG. **2A**, the X-rail **171** is coupled with and slides along the Y-rail **172**, and the Z-rail **173** is coupled with and slides along the X-rail **171**; however, in other variations, the X-rail **171**, Y-rail **172**, and Z-rail **173** can be coupled and interact with each other in another suitable manner. In the variation shown in FIG. **2A**, the gantry **170** is configured to move tools, using the X-rail **171** and the Y-rail **172**, into alignment with elements of the deck no, and is configured to move tools, using the Z-rail **173** for accessing elements of the deck (e.g., portions of the reagent cartridge **120**, elements of the sample processing cartridge **130**, tools of the tool container **140**, etc.). In some variations, the gantry **170** can be configured to return to a parked position (e.g., once a protocol is completed, when the system is turned off, before the lid is opened, etc.), where the parked position allows the user to access appropriate areas of the deck (e.g., for refilling tips, filling tubes with reagents, removing a microwell cartridge, etc.). However, the gantry **170** can have another suitable baseline position.

Various operation modes of the gantry **170** are further described in relation to workflows of Section 3 below.

(146) 2.2.1 Gantry—Pipettor

(147) As shown in FIG. 2A, the gantry **170** can include and/or or be configured to interact with a pipettor **174**, which functions to hold, move, and/or otherwise interact with any number of tips or other tools, such as those of the tool container **140** described above. In variations, the pipettor **174** assembly can include one or more of: a pump (e.g., displacement pump) for providing pressure differentials for delivery and aspiration of fluids, a pressure sensor for sensing pipetting pressure, a level sensor for sensing fluid level within the pipettor **174**, a tip detector (e.g., to enable determination of presence or absence of a tip coupled to the pipettor **174**), and a tip ejection motor coupled to a tip ejector for removing tips from the pipettor **174**. As shown in FIG. 2A, the pipettor **174** can be coupled to the Z-rail **173** of the gantry **170**; however, in other variations, the pipettor **174** can additionally or alternatively be coupled to other portions of the gantry **170**.

(148) The pipettor **174** is preferably operable in an automated fashion (e.g., motorized, mechanized, self-operating, etc.) and can be configured to control any or all of the following predetermined parameters: volume (e.g. dispensing exact volumes, aspirating exact volumes), a height above the well at which each material is dispensed (e.g. priming buffer is dispensed between 0.25 and 0.3 millimeters above the top of each well, cell suspension is dispensed at a height of 0.25 millimeters above the top of each well, etc.), or can control any other suitable property according to any suitable parameter. Additionally or alternatively, the pipettor **174** can be configured to operate in a manual fashion (e.g., according to a user, with user intervention, held and used by a user, etc.) or in any suitable way. In yet another embodiment, the pipettor **174** may be used to pick up one or more tools associated with the tool container, such as any or all of: a mechanical tool, magnetic tool, an optical tool, and any other suitable tool. The tools can be moved by the pipettor **174** to the reagent cartridge and/or the microwell cartridge such that the tool(s) can perform specific mechanical/magnetic and/or optical functions with respect to specific contents of the reagent cartridge or microwell cartridge.

(149) 2.2.2 Gantry—Other Tools

(150) Additionally, the gantry **170** can include or support one or more tools described in relation to other elements above. For instance, the gantry **170** can be coupled with a unit of lid-opening tool **145** (shown in FIG. 2B) for transitioning the lid **135** of the sample processing cartridge **130** between open and closed states. The gantry **170** can additionally or alternatively be coupled with a magnetic head actuator **175** (shown in FIG. 2B) associated with the separation subsystem **160** described above (e.g., in relation to actuation of the first body **161** with magnetic distal region **163**) for separation of target material from non-target material.

(151) In some variations, the gantry **170** can directly or indirectly be coupled with a camera **176** (e.g., camera coupled with a light), which functions to enable reading of tags (e.g., barcodes) associated with various disposables supported by the deck **110** (e.g., in relation to identifying proper setup of a run, in relation to inventory management, etc.). Additionally or alternatively, the camera **176** can include functionality for transmitting image data capturing configurations of elements at the deck no without reading of specific tags. As shown in FIGS. 2A and 2F, the camera **176** can be coupled to the Z-rail **173** with pipettor **174** described above, such that the camera **176** has a field of view associated with objects that the pipettor **174** is aligned with. However, motion of the camera **176** can alternatively be uncoupled from the pipettor **174**, such that the camera **176** can be moved independently of the pipettor **174**. In variations, the camera **176** can include components described in applications incorporated by reference above. The camera image can also be used as an additional feedback mechanism for the precise movement of the gantry **170** and/or its components (e.g., pipettor, tools, etc.) to reach a desired location, thereby dramatically improving the accuracy and precision of the motion system (e.g., to less than 100 microns, less than 50 microns, less than 25 microns, less than 10 microns, less than 5 microns, less than 1 micron, etc.).

(152) Additionally or alternatively, as shown in FIGS. 1A and 1B, the gantry **170** can include various sensors for monitoring environmental or other parameters of the system **100**. For instance, in variations, the sensors can include one or more of: temperature sensors, humidity sensors, pressure sensors, optical sensors, vibration sensors, and other suitable sensors. The sensors can, however, be positioned relative to the system **100** in another suitable manner (e.g., uncoupled from the gantry **170**, positioned at the deck **110**, positioned at the base **180**, etc.).

(153) 2.3 System—Base

(154) As shown in FIG. 1A, the system **100** can include a base **180**, which functions to support control and processing architecture associated with elements of the deck **110** and gantry **170** described above. In variations, the base **180** can support control and processing architecture for one or more system functions including: pressure modification for fluid delivery throughout the sample processing cartridge **130** and/or pipettor **174**; fluid level sensing (e.g., at the sample processing cartridge **130**, at the pipettor **174**, at various storage volumes of the reagent cartridge **120**, etc.); actuation of lid opening mechanisms of the sample processing cartridge **130**; thermocycling and/or other heating functions for the reagent cartridge **120** and/or sample processing cartridge **130**; cooling functions for the reagent cartridge **120** and/or sample processing cartridge **130**; separation functions (e.g., elution, magnetic separation, other separation, etc.); functions for control of the gantry **170**; functions involving receiving sensor signals and returning outputs; functions involving receiving sensor signals and executing various actions; functions for transitioning system doors between various states (e.g., open states, closed states, locked states, unlocked states, etc.); functions associated with system power management; functions associated with system status indication elements (e.g., lights, audio output devices, visual output devices, etc.); functions associated with system input devices (e.g., buttons, keyboards, keypads, mice, joysticks, switches, touch screens, etc.); functions associated with display devices; functions associated with system data storage devices; functions associated with system transmission devices (e.g., wired transmission devices, wireless transmission devices, etc.); and other suitable functions.

(155) In variations, the base **180** can thus support an electronics subsystem (e.g. PCB, power source, communication module, encoder, etc.) associated with a processing architecture (e.g. onboard the system, separate from the system, etc.), or any other suitable component, where the processing architecture can include any or all of: processors (e.g. microprocessors), controllers (e.g. microcontrollers), memory, storage, software, firmware, or any other suitable component. Additionally, the processing subsystem can include a machine vision module, which functions to read tags, verify protocols, perform error detection (e.g. detect that reagents do not match an assigned protocol), or perform any other function.

(156) For instance, in an example operation flow, an operator can initiate the performance of the protocol (e.g., by pushing a button of the system, by interacting a touch-sensitive display of the system to make a selection, etc.). A barcode reader performs an error detection protocol by scanning tags of the deck elements (e.g, reagent cartridge, sample processing cartridge, tool container, etc.) and comparing with the protocol selected by the user; if the tags do not match the selected protocol, a notification can be transmitted to the user, and if the tags are correct, the protocol can begin. At this point, the operator may no longer needed. According to one or more workflows, some of which are described in Section 3 below, the correct types and volumes of materials (e.g., reagents/samples) are added to or removed from the sample processing cartridge at the correct times in an automated fashion. Once the protocol is complete, the operator can proceed with collecting and/or processing the contents of the microwell cartridge as desired, and/or setting up a new run. Variations of methods and workflows enabled by the system **100** are further described below.

(157) Embodiments, variations, and examples of control and processing architecture are further described in U.S. application Ser. No. 16/048,104, filed 27 Jul. 2018; U.S. application Ser. No. 16/049,057, filed 30 Jul. 2018; U.S. application Ser. No. 15/720,194, filed 29 Sep. 2017; U.S.

application Ser. No. 15/430,833, filed 13 Feb. 2017; U.S. application Ser. No. 15/821,329, filed 22 Nov. 2017; U.S. application Ser. No. 15/782,270, filed 12 Oct. 2017; U.S. application Ser. No. 16/049,240, filed 30 Jul. 2018; U.S. application Ser. No. 15/815,532, filed 16 Nov. 2017; U.S. application Ser. No. 16/115,370, filed 28 Aug. 2018, U.S. application Ser. No. 16/564,375, filed 9 Sep. 2019, and U.S. application Ser. No. 16/816,817, filed 12 Mar. 2020, as incorporated by reference above.

(158) 2.4 System—Functional Coupling Between Deck, Gantry, and Base

(159) In variations, the deck **110**, gantry **170**, and base **180** of the system can be functionally coupled to provide various functions. Functional coupling can be electromechanical, mechanical, electrical, magnetic, pneumatic, fluidic, optical, and/or of other coupling means, according to various modes of operation. FIGS. **14A-14C** depict functional coupling for a variation of a system, as described below.

(160) FIG. **14A** depicts an example of mechanical, electrical, pneumatic, optical, and magnetic communication between various elements of the deck **110**, gantry **170**, and base **180**. In more detail, as shown in FIG. **14A**, the gantry **170** supports actuation-associated elements including a Z-axis arm **801** associated with Z-rail **173** described above, 3-axis gantry motors **802**, 3-axis encoders **803**, 3-axis optical switches **804**, a magnetic actuator **805**, and a lid-opening tool **806**. The gantry **170** further supports pipetting elements including: a pipettor **174**, a displacement pump **808**, a pipette tip detector **809**, a pipette tip ejection motor **810**, a level sensor **811**, and a pressure sensor **812**. The gantry **170** also supports a camera and light **176** and a set of sensors **814** including temperature and humidity sensors. As shown in FIG. **14A**, the deck **110** supports a sample processing cartridge **130**, a reagent cartridge **120**, and a tool container **140** including a set of tools **865**. As shown in FIG. **14A**, the base **180** supports a first computing subsystem **815**, one or more interface boards **816**, and gantry-control circuitry **817**. Mechanical, electrical, pneumatic, optical, and magnetic communication between various elements of the deck **110**, gantry **170**, and base **180** are depicted with various line types in FIG. **14A**.

(161) FIG. **14B** depicts an example of mechanical, fluidic, magnetic, and pneumatic communication between various elements of the deck **110** and base **180**. In more detail, as shown in FIG. **14B**, the deck **110** supports a reagent cartridge **120** including: a storage volume for ambient reagents and functional beads **818**, a sample storage volume **819** for receiving a raw sample **820**, a storage volume for storing chilled reagents and/or results of sample processing **821**, a set of storage volumes for separation particles and associated separation reservoirs **822**, and a set of storage volumes for PCR **823**, with associated PCR covers **824**; a tool container **140** including a set of tools (e.g., pipette tips) **825**; a sample processing cartridge **130**; and a set of deck tools **826** including a heater head **827**, a suction head **828**, and a magnetic head **829**. As shown in FIG. **14B**, a set of elements interface the deck **110** with the base **180**, where the set of elements can include one or more of: a vacuum port **830**, a pinch actuator **831**, a sample processing cartridge thermal body **832** (configured to mate with the microwell region of the sample processing cartridge **130** or other region of the sample processing cartridge **130**), a first reagent cartridge thermal body **833** (configured to mate with a domain of the reagent cartridge **120**, at the first region in, for chilled reagents), a second reagent cartridge thermal body **834** (configured to mate with a domain of the reagent cartridge **120**, at the first region in, for PCR thermocycling), a fluid level sensor **836**, and a magnet **835** for separation processes. Mechanical, fluidic, magnetic, and pneumatic communication between various elements of the deck **110** and base **180** are depicted with various line types in FIG. **14B**.

(162) FIG. **14C** depicts an example of electrical communication between various elements of the deck **110** and base **180**. In more detail, as shown in FIG. **14C**, the deck **110** supports a sample processing cartridge **130**; a reagent cartridge **120** including: a storage volume for storing chilled reagents and/or results of sample processing **821**, a set of storage volumes for separation particles and associated separation reservoirs **822**, and a set of storage volumes for PCR **823** with associated

PCR covers; and a set of deck tools **826** including a heater head **827**. As shown in FIG. **14C**, a set of elements interface the deck **110** with the base **180**, where the set of elements can include one or more of: a vacuum port **830**, a pinch actuator **831**, a sample processing cartridge thermal body **832**, a first reagent cartridge thermal body **833**, a second reagent cartridge thermal body **834**, a magnet **835** for separation processes, and a fluid level sensor **836**.

(163) In more detail, the vacuum port **830** is coupled to a pumping subsystem **837** which includes a vent valve manifold **838**, and a pressure sensor **839**. The sample processing cartridge thermal body **832** is coupled to a thermocycler **840** of a heating and cooling subsystem, which includes a heater **841** (e.g., Peltier heater, thermocouples, heat sinks, and fans) and thermal control circuitry **842**. The first reagent cartridge thermal body **833** is coupled to a cooling device **843** of a heating and cooling subsystem, which includes a cooling element **844** (e.g., Peltier cooler, thermocouples, heat sinks, and fans) and thermal control circuitry **845**. The second reagent cartridge thermal body **834** is coupled to a heating device (e.g., for on-board PCR) of a heating and cooling subsystem, which includes a first heater **846** (e.g., Peltier heater, thermocouples, heat sinks, and fans), a second heater **847** coupled to the heat sink, and thermal control circuitry **848**. The magnet **835** is coupled to a magnet actuator **849**. Finally, the fluid level sensor **836** is coupled to a fluid level controller **850**.

(164) As shown in FIG. **14C**, the base **180** further supports a first computing subsystem **815**, one or more interface boards **816**, a speaker **851**, a door locking element **852** (interfacing with a door magnet **853** to provide operation modes of a door **90**, as described above), a power inlet **854** coupled to a power supply **855** coupled to AC mains **856**, a power switch **857**, indicator lights **858** (e.g., for power/ground effects), exhaust fans **859**, a set of ports (e.g., including an ethernet port **860** and USB ports **861**), and a display **862** (e.g., touchscreen display). Elements of the base can further interface with off-system accessories (e.g., input devices **863**, storage devices **864**).

Electrical, communication between various elements of the deck **110** and base **180** are depicted with various line types in FIG. **14C**.

(165) 3. Method

(166) As shown in FIG. **15**, an embodiment of a method **300** for sample processing includes: positioning a reagent cartridge, a tool container, and a sample processing cartridge at a deck of a sample processing system **S310**; transmitting content of the reagent cartridge between the sample processing cartridge by using tools of the tool container in an operational sequence **S320**; and processing a sample at the sample processing cartridge and/or reagent cartridge **S330**. Additionally or alternatively, the method **300** can include any or all of the processes described in U.S.

application Ser. No. 16/048,104, filed 27 Jul. 2018; U.S. application Ser. No. 16/049,057, filed 30 Jul. 2018; U.S. application Ser. No. 15/720,194, filed 29 Sep. 2017; U.S. application Ser. No. 15/430,833, filed 13 Feb. 2017; U.S. application Ser. No. 15/821,329, filed 22 Nov. 2017; U.S. application Ser. No. 15/782,270, filed 12 Oct. 2017; U.S. application Ser. No. 16/049,240, filed 30 Jul. 2018; U.S. application Ser. No. 15/815,532, filed 16 Nov. 2017; U.S. application Ser. No. 16/115,370, filed 28 Aug. 2018, U.S. application Ser. No. 16/564,375, filed 9 Sep. 2019, and U.S. application Ser. No. 16/816,817, filed 12 Mar. 2020, which are each incorporated in their entirety by this reference.

(167) The method is preferably performed with an embodiment, variation, or example of the systems described above (e.g., in relation to transmission of content between various elements and/or sample processing), but can additionally or alternatively be performed with any other suitable system. The method is further preferably at least partially automated (e.g., requires user to load reagents and select a protocol, requires no user intervention, etc.), but one or more portions can additionally or alternatively be manually performed (e.g., for quality control steps, for all protocols, for rare protocols, etc.).

(168) Specific workflows associated with the method **300** and system elements described above are described in further detail below, where samples (e.g., samples including cell-derived material, proteins, mRNAs, proteins and mRNA; samples that include multiple samples each tagged with

multiplexing barcodes; samples that include encapsulated particles from either cell or non-cell derived biomarkers, etc.) can be processed according to the workflows (e.g., workflows in Sections 3.1-3.3 below), followed by library preparation workflows (e.g., workflow in Section 3.4 below), followed by next generation sequencing (NGS).

(169) 3.1 Method—Example Workflow for a 3' Protocol for mRNA Synthesis/cDNA Amplification
(170) As shown in FIG. 16, a variation of the method, configured for mRNA synthesis/cDNA amplification **300'** can include: performing a run preparation operation, wherein the run preparation operation configures a system for performing an mRNA synthesis/cDNA amplification protocol **S305'**; priming a sample processing cartridge of the system upon completion of the run preparation operation **S310'**; co-capturing a set of cells of a sample with a set of functionalized particles, in single cell format, at the sample processing cartridge **S315'**; lysing the set of cells followed by binding released mRNA from the lysed cells with the functionalized particles (e.g., barcoded microspheres) (e.g., with associated washing the functionalized particles of any unbound mRNA) **S320'**; performing a reverse transcription operation; retrieving the set of functionalized particles, with associated bound target content, from the sample processing cartridge **S325'**; performing an exonuclease treatment operation with a volume of cell-derived content associated with the set of functionalized particles **S330'**; performing a strand denaturing and second strand synthesis operation with the volume of cell-derived content **S335'**; performing a cDNA amplification operation with the volume of cell-derived content **S340'**; performing an mRNA particle purification operation with PCR product of the cDNA amplification operation **S345'**; and performing a run completion operation upon completion of the mRNA particle purification operation **S360'**.

(171) In more detail, performing a run preparation operation, **S305'** can include sub-steps associated with one or more of: preparing a cell suspension; initializing and performing operational checks of system subsystems (e.g., associated with the deck, associated with the gantry, associated with the base, etc.); returning the gantry to a home position; removing one or more seals from the reagent cartridge and/or loading reagents onto the reagent cartridge; positioning a sample processing cartridge unit; removing one or more seals from the tool container positioned at the deck; dispensing the cell suspension into a storage container prior to use; verifying proper positioning and states (e.g., in relation to expiration dates) of disposables for the protocol, upon scanning tags of disposables with a camera (e.g., machine vision camera); receiving sample identification information (e.g., from an operator); and initiating run of the sample. Steps of **S305'** can be implemented by the system automatically and/or by an operator. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(172) In more detail, priming a sample processing cartridge of the system upon completion of the run preparation operation **S310'** and co-capturing a set of cells of a sample with a set of functionalized particles, in single cell format, at the sample processing cartridge **S315'** can include one or more of: dispensing a priming solution (e.g., in a manner that prevents bubbles from being trapped within the sample processing cartridge) into the inlet reservoir of a sample processing cartridge; incubating the priming solution within the sample processing cartridge; dispensing one or more wash solutions into the inlet reservoir of the sample processing cartridge; transmitting solutions to a waste containment region of the sample processing cartridge; dispensing a cell suspension into the inlet reservoir of the sample processing cartridge and capturing cells, in single-cell format, within wells of the sample processing cartridge; dispensing a set of functionalized particles into the inlet reservoir of the sample processing cartridge and co-capturing the set of functionalized particles with the set of cells; incubating content of the wells of the sample processing cartridge; and picking up/releasing various tools (e.g., by a gantry coupled to a pipette interface) involved with the substep(s). Steps **S310'** and **S315'** are preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(173) In more detail, performing a reverse transcription operation, at the sample processing

cartridge, with lysate of the set of cells **S320'** in the presence of functionalized particles (e.g., barcoded microspheres) can include one or more of: dispensing one or more wash solutions into the inlet reservoir of the sample processing cartridge; transmitting solutions to a waste containment region of the sample processing cartridge; dispensing a particle-binding buffer into the inlet reservoir of the sample processing cartridge; dispensing a DTT solution into the inlet reservoir of the sample processing cartridge; dispensing a lysis solution into the inlet reservoir of the sample processing cartridge; displacing fluid above wells of the sample processing cartridge with an oil, thereby isolating contents of wells and preventing undesired material transfer across wells (e.g., as in U.S. application Ser. No. 16/564,375 filed 9 Sep. 2019, which is herein incorporated in its entirety by this reference); displacing the oil with air from the inlet reservoir; dispensing a particle-binding wash buffer into the inlet reservoir of the sample processing cartridge; dispensing a pre-RT reaction wash buffer into the inlet reservoir of the sample processing cartridge; dispensing a cDNA synthesis solution (e.g., SuperScript IV™) into the inlet reservoir of the sample processing cartridge; dispensing an RT cocktail into the inlet reservoir of the sample processing cartridge; incubating contents of the sample processing cartridge; performing incubation steps; and picking up/releasing various tools involved with the substep(s). Step **S320'** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(174) In more detail, retrieving the set of functionalized particles, with associated bound target content, from the sample processing cartridge **S325'** can include performing magnetic separation operations (e.g., as described above), using manual or automatic operations. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(175) In more detail, performing an exonuclease treatment operation with a volume of cell-derived content associated with the set of functionalized particles **S330'** can include one or more of: mixing water and exonuclease buffer to produce an exonuclease solution having a desired concentration; dispensing the exonuclease solution, with functionalized particles into a first PCR container; dispensing an oil (e.g., mineral oil) into the first PCR container; thermocycling and incubating contents of the first PCR container; extracting a product of the first PCR container; performing a separation operation with the product of the first PCR container; discarding waste from the separation operation; performing incubation steps; and picking up/releasing various tools involved with the substep(s). Step **S330'** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(176) In more detail, performing a strand denaturing and second strand synthesis operation **S335'** can include one or more of: transferring a hydroxide solution (e.g., sodium hydroxide solution) to a second magnetic separation container; mixing contents of the second magnetic separation container; activating a magnetic separation subsystem (e.g., set of magnets coupled to actuator, described above) in proximity to the second magnetic separation container, thereby separating functionalized magnetic particles toward the magnet(s); discarding waste material from the second magnetic separation container; dispensing a wash solution to the second magnetic separation container; mixing a second strand synthesis primer enzyme within a process container in proximity to the second magnetic separation container; mixing the second strand synthesis primer enzyme with contents of the second magnetic separation container; dispensing product of the second magnetic separation container into a second PCR container; thermocycling contents of the second PCR container, with mixing; transferring product of the second PCR container to a third magnetic separation container; magnetically separating product of the third magnetic separation container from waste and discarding waste; transferring a wash solution to the third magnetic separation container; and picking up/releasing various tools involved with the substep(s). Step **S335'** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as

recommended.

(177) In more detail, performing a cDNA synthesis operation **S340'** can include one or more of: mixing a polymerase blend (e.g., Kapa Biosystems HiFi Hotstart Ready Mix™) within a cold storage volume; mixing the PCR master mix from the cold storage volume with contents of the third magnetic separation container; aliquoting contents of the third magnetic separation container into a third, fourth, fifth, and sixth PCR operation containers; aliquoting an oil (e.g., mineral oil) into the third, fourth, fifth, and sixth PCR operation containers; running a third PCR operation; and picking up/releasing various tools involved with the substep(s). Step **S340'** is preferably performed automatically by the system but can alternatively be performed in another suitable manner.

Furthermore, various sub-steps can be performed once, or repeated as recommended.

(178) In more detail, performing an mRNA particle purification operation with PCR product of the cDNA amplification operation **S345'** can include one or more of: dispensing product from the third, fourth, fifth, and sixth PCR operation containers into a fourth magnetic separation container; mixing and transferring PCR purification particles (e.g., AMPure beads XP™) into the fourth magnetic separation container; diluting and mixing ethanol with nuclease-free water within a tenth magnetic separation container; removing waste from the tenth magnetic separation container, after incubation; mixing nuclease-free water with target content of the tenth magnetic separation container; magnetically separating target mRNA-cDNA product from tenth magnetic separation container; transferring target mRNA-cDNA product from tenth magnetic separation container to cold storage; and picking up/releasing various tools involved with the substep(s). Step **S345'** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(179) In more detail, performing a run completion operation upon completion of the mRNA particle purification operation **S360'** can include one or more of: returning the gantry to a home configuration; providing a notification that sample processing is complete; releasing the reagent cartridge and/or sample processing cartridge from the system for off-board storage; discarding system waste; performing a system cleaning operation; and transitioning the system to a deactivated (e.g., idle, off) state. Steps of **S360'** can be implemented by the system automatically and/or by an operator. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(180) Example details of steps of the method **300'** are further described in TABLE 1 of Appendix A. In relation to steps of the method **300'**, descriptions of ambient temperature and chilled reagents, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 2 of Appendix A. In relation to TABLE 2 of Appendix A, volumes of reagents can be scaled according to sizes of sample processing chips used and/or number of reactions run. In relation to magnetic separation operations, descriptions of apparatus and associated reagents used, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 3 of Appendix A. In relation to amplification (e.g., PCR) operations, descriptions of apparatus and associated reagents used, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 4 of Appendix A. In relation to fluid aspiration and delivery operations, descriptions of apparatus, as well as positions associated with an embodiment of the tool container described above are described in TABLE 5 of Appendix A. In relation to actuation of components for automation of protocol aspects, gantry arm and pipettor operations (e.g., in relation to apparatus coupling with disposables, apparatus uncoupling from disposables, fluid mixing, waste discarding, aspiration, delivery, aliquoting, etc.) are described in TABLE 6 of Appendix A. In relation to transitioning between modes for automation of protocol aspects, sample processing cartridge operations are described in TABLE 7 of Appendix A. In relation to transitioning between modes for automation of protocol aspects, heating and cooling

subsystem operation modes are described in TABLE 8 of Appendix A. In relation to transitioning between modes for automation of protocol aspects, magnetic separation subsystem operations are described in TABLE 9 of Appendix A. In relation to amplification operations, PCR program details associated with the method **300'** are described in TABLE 10 of Appendix A.

(181) 3.2 Method—Example Workflow for Single Cell Cytometry

(182) As shown in FIG. 17, a variation of the method, configured for single cell cytometry **300''** can include: performing a run preparation operation, wherein the run preparation operation configures a system for performing a single cell cytometry protocol **S305''**; priming a sample processing cartridge of the system upon completion of the run preparation operation **S310''**; co-capturing a set of cells of a sample with a set of functionalized particles, in single cell format, at the sample processing cartridge **S315''**; lysing the set of cells followed by binding released antibody-tagged oligonucleotides from the lysed cells to functionalized particles (e.g., barcoded microspheres) **S320''** (e.g., with washing the unbound materials and then performing a reverse transcription operation); retrieving the set of functionalized particles, with associated bound target content, from the sample processing cartridge **S325''**; performing an exonuclease treatment operation with a volume of cell-derived content associated with the set of functionalized particles **S330''**; performing a strand denaturing and second strand synthesis operation with the volume of cell-derived content **S335''**; performing a cDNA amplification operation with the volume of cell-derived content **S340''**; performing a particle purification operation with PCR product of the cDNA amplification operation **S345''**; performing a set of antibody-derived tag purification and amplification operations with outputs of the particle purification operation **S350''**; performing a set of mRNA purification and amplification operations with outputs of the set of antibody-derived tag purification and amplification operations **S355''**; and performing a run completion operation upon completion of the set of mRNA purification and amplification operations **S360''**.

(183) In more detail, performing a run preparation operation, **S305''** can include sub-steps associated with one or more of: preparing a cell suspension; initializing and performing operational checks of system subsystems (e.g., associated with the deck, associated with the gantry, associated with the base, etc.); returning the gantry to a home position; removing one or more seals from the reagent cartridge and/or loading reagents onto the reagent cartridge; positioning a sample processing cartridge unit; removing one or more seals from the tool container positioned at the deck; dispensing the cell suspension into a storage container prior to use; verifying proper positioning and states (e.g., in relation to expiration dates) of disposables for the protocol, upon scanning tags of disposables with a camera (e.g., machine vision camera); receiving sample identification information (e.g., from an operator); and initiating run of the sample. Steps of **S305''** can be implemented by the system automatically and/or by an operator. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(184) In more detail, priming a sample processing cartridge of the system upon completion of the run preparation operation **S310''** and co-capturing a set of cells of a sample with a set of functionalized particles, in single cell format, at the sample processing cartridge **S315''** can include one or more of: dispensing a priming solution into the inlet reservoir of a sample processing cartridge; incubating the priming solution within the sample processing cartridge; dispensing one or more wash solutions into the inlet reservoir of the sample processing cartridge; transmitting solutions to a waste containment region of the sample processing cartridge; dispensing a cell suspension into the inlet reservoir of the sample processing cartridge and capturing cells, in single-cell format, within wells of the sample processing cartridge; dispensing a set of functionalized particles into the inlet reservoir of the sample processing cartridge and co-capturing the set of functionalized particles with the set of cells; incubating content of the wells of the sample processing cartridge; and picking up/releasing various tools involved with the substep(s). Steps **S310''** and **S315''** are preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or

repeated as recommended.

(185) In more detail, performing a reverse transcription operation, at the sample processing cartridge, with lysis of the set of cells **S320"** can include one or more of: dispensing one or more wash solutions into the inlet reservoir of the sample processing cartridge; transmitting solutions to a waste containment region of the sample processing cartridge; dispensing a particle-binding buffer into the inlet reservoir of the sample processing cartridge; dispensing a DTT solution into the inlet reservoir of the sample processing cartridge; dispensing a lysis solution into the inlet reservoir of the sample processing cartridge; displacing fluid above wells of the sample processing cartridge with an oil, thereby isolating contents of wells and preventing undesired material transfer across wells (e.g., as in U.S. application Ser. No. 16/564,375 filed 9 Sep. 2019, which is herein incorporated in its entirety by this reference); displacing the oil with air from the inlet reservoir; dispensing a particle-binding wash buffer into the inlet reservoir of the sample processing cartridge; dispensing a pre-RT reaction wash buffer into the inlet reservoir of the sample processing cartridge; dispensing a cDNA synthesis solution (e.g., SuperScript IV™) into the inlet reservoir of the sample processing cartridge; dispensing an RT cocktail into the inlet reservoir of the sample processing cartridge; incubating contents of the sample processing cartridge; performing incubation steps; and picking up/releasing various tools involved with the substep(s). Step **S320"** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(186) In more detail, retrieving the set of functionalized particles, with associated bound target content, from the sample processing cartridge **S325"** can include performing magnetic separation operations (e.g., as described above), using manual or automatic operations. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(187) In more detail, performing an exonuclease treatment operation with a volume of cell-derived content associated with the set of functionalized particles **S330"** can include one or more of: mixing water and exonuclease buffer to produce an exonuclease solution having a desired concentration; dispensing the exonuclease solution, with functionalized particles into a first PCR container; dispensing an oil (e.g., mineral oil) into the first PCR container; thermocycling and incubating contents of the first PCR container; extracting a product of the first PCR container; performing a separation operation with the product of the first PCR container; discarding waste from the separation operation; performing incubation steps; and picking up/releasing various tools involved with the substep(s). Step **S330"** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(188) In more detail, performing a strand denaturing and second strand synthesis operation **S335"** can include one or more of: transferring a hydroxide solution (e.g., sodium hydroxide solution) to a second magnetic separation container; mixing contents of the second magnetic separation container; activating a magnetic separation subsystem (e.g., set of magnets coupled to actuator, described above) in proximity to the second magnetic separation container, thereby separating functionalized magnetic particles toward the magnet(s); discarding waste material from the second magnetic separation container; dispensing a wash solution to the magnetic separation container; mixing a second strand synthesis primer enzyme within a process container in proximity to the second magnetic separation container; mixing the second strand synthesis primer enzyme with contents of the second magnetic separation container; dispensing product of the second magnetic separation container into a second PCR container; thermocycling contents of the second PCR container, with mixing; transferring product of the second PCR container to a third magnetic separation container; magnetically separating product of the third magnetic separation container from waste and discarding waste; transferring a wash solution to the third magnetic separation container; and picking up/releasing various tools involved with the substep(s). Step **S335"** is preferably performed automatically by the system but can alternatively be performed in another

suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(189) In more detail, performing a cDNA synthesis operation S340" can include one or more of: mixing a polymerase blend (e.g., Kapa Biosystems HiFi Hotstart Ready Mix™) within a cold storage volume; mixing the PCR master mix from the cold storage volume with contents of the third magnetic separation container; aliquoting contents of the third magnetic separation container into a third, fourth, fifth, and sixth PCR operation containers; aliquoting an oil (e.g., mineral oil) into the third, fourth, fifth, and sixth PCR operation containers; running a first PCR operation; and picking up/releasing various tools involved with the substep(s). Step S340" is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(190) In more detail, performing a particle purification operation with PCR product of the cDNA amplification operation S345" can include one or more of: dispensing product from the third, fourth, fifth, and sixth PCR operation containers into a fourth magnetic separation container; separating target content of the fourth magnetic separation container and transmitting target content into a fifth magnetic separation container; mixing and transferring PCR purification particles (e.g., AMPure beads XP™) into the fifth magnetic separation container; transferring product of the fifth magnetic separation container into a sixth separation container; and picking up/releasing various tools involved with the substep(s). Step S345" is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(191) In more detail, performing a set of antibody-derived tag purification and amplification operations with outputs of the particle purification operation S350" can include one or more of: mixing and transferring PCR purification particles (e.g., AMPure beads XP™) into the sixth magnetic separation container; discarding waste from the sixth magnetic separation container; diluting and mixing ethanol with nuclease-free water within a process container; transferring ethanol to the sixth magnetic separation container; discarding waste from the sixth magnetic separation container, upon performing magnetic separation; transferring target content of the sixth magnetic separation container into a seventh magnetic separation container; mixing PCR purification particles (e.g., AMPure beads XP™) into the seventh magnetic separation container; transferring ethanol to the seventh magnetic separation container; discarding waste from the seventh magnetic separation container, upon performing magnetic separation; dispensing water into the seventh magnetic separation container; transferring purified cDNA from the seventh magnetic separation container into a seventh PCR operation container; and picking up/releasing various tools involved with the substep(s). Step S350" is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(192) In more detail, performing a set of mRNA purification and amplification operations with outputs of the set of antibody-derived tag purification and amplification operations S355" can include one or more of: transferring PCR indexing primers into the seventh PCR operation container; transferring a polymerase blend into the seventh PCR operation container; mixing contents of the seventh PCR operation container; transferring an oil (e.g., mineral oil) into the seventh PCR operation container; initiating a second PCR operation; transferring PCR product from the seventh PCR operation container to an eighth magnetic separation container; transferring PCR purification particles (e.g., AMPure beads XP) into the eighth magnetic separation container; transferring ethanol to the eighth magnetic separation container; discarding waste from the eighth magnetic separation container, upon performing magnetic separation; dispensing water into the eighth magnetic separation container; transferring purified cDNA from the eighth magnetic separation container into a storage container; and picking up/releasing various tools involved with the substep(s). Step S355" is preferably performed automatically by the system but can

alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(193) In more detail, performing a run completion operation upon completion of the set of mRNA purification and amplification operations **S360''** can include one or more of: returning the gantry to a home configuration; providing a notification that sample processing is complete; releasing the reagent cartridge and/or sample processing cartridge from the system for off-board storage; discarding system waste; performing a system cleaning operation; and transitioning the system to a deactivated (e.g., idle, off) state. Steps of **S360''** can be implemented by the system automatically and/or by an operator. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(194) Example details of steps of the method **300''** are further described in TABLE 1 of Appendix B. In relation to steps of the method **300''**, descriptions of ambient temperature and chilled reagents, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 2 of Appendix B. In relation to TABLE 2 of Appendix B, volumes of reagents can be scaled according to sizes of sample processing chips used and/or number of reactions run. In relation to magnetic separation operations, descriptions of apparatus and associated reagents used, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 3 of Appendix B. In relation to amplification (e.g., PCR) operations, descriptions of apparatus and associated reagents used, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 4 of Appendix B. In relation to fluid aspiration and delivery operations, descriptions of apparatus, as well as positions associated with an embodiment of the tool container described above are described in TABLE 5 of Appendix B. In relation to actuation of components for automation of protocol aspects, gantry arm and pipettor operations (e.g., in relation to apparatus coupling with disposables, apparatus uncoupling from disposables, fluid mixing, waste discarding, aspiration, delivery, aliquoting, etc.) are described in TABLE 6 of Appendix B. In relation to transitioning between modes for automation of protocol aspects, sample processing cartridge operations are described in TABLE 7 of Appendix B. In relation to transitioning between modes for automation of protocol aspects, heating and cooling subsystem operation modes are described in TABLE 8 of Appendix B. In relation to transitioning between modes for automation of protocol aspects, magnetic separation subsystem operations are described in TABLE 9 of Appendix B. In relation to amplification operations, PCR program details associated with the method **300''** are described in TABLE 10 of Appendix B.

(195) 3.3 Method—Example Workflow for CITE-Seq Protocol (Single Cell Multiomics)

(196) As shown in FIG. 18, a variation of the method, configured for single cell multiomics **300'''** can include: performing a run preparation operation, wherein the run preparation operation configures a system for performing a single cell multiomics protocol **S305'''**; priming a sample processing cartridge of the system upon completion of the run preparation operation **S310'''**; co-capturing a set of cells of a sample with a set of functionalized particles, in single cell format, at the sample processing cartridge **S315'''**; at the sample processing cartridge, with lysis of the set of cells, binding mRNA and antibody tagged oligos from the lysed cells to the set of functionalized particles, with washing of unbound materials followed by performing a reverse transcription reaction **S320'''**; retrieving the set of functionalized particles, with associated bound target content, from the sample processing cartridge **S325'''**; performing an exonuclease treatment operation with a volume of cell-derived content associated with the set of functionalized particles **S330'''**; performing a strand denaturing and second strand synthesis operation with the volume of cell-derived content **S335'''**; performing a cDNA amplification operation with the volume of cell-derived content **S340'''**; performing a particle purification operation with PCR product of the cDNA amplification operation **S345'**; performing a set of antibody-derived tag purification and amplification operations with outputs of the particle purification operation **S350'''**; performing a set

of purification and amplification operations with outputs of the set of antibody-derived tag purification and amplification operations S355'''; performing a set of mRNA particle purification and amplification operations S360'''; and performing a run completion operation upon completion of the set of mRNA purification and amplification operations S35'''.

(197) In more detail, performing a run preparation operation, S305''' can include sub-steps associated with one or more of: preparing a cell suspension; initializing and performing operational checks of system subsystems (e.g., associated with the deck, associated with the gantry, associated with the base, etc.); returning the gantry to a home position; removing one or more seals from the reagent cartridge and/or loading reagents onto the reagent cartridge; positioning a sample processing cartridge unit; removing one or more seals from the tool container positioned at the deck; dispensing the cell suspension into a storage container prior to use; verifying proper positioning and states (e.g., in relation to expiration dates) of disposables for the protocol, upon scanning tags of disposables with a camera (e.g., machine vision camera); receiving sample identification information (e.g., from an operator); and initiating run of the sample. Steps of S305''' can be implemented by the system automatically and/or by an operator. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(198) In more detail, priming a sample processing cartridge of the system upon completion of the run preparation operation S310''' and co-capturing a set of cells of a sample with a set of functionalized particles, in single cell format, at the sample processing cartridge S315''' can include one or more of: dispensing a priming solution into the inlet reservoir of a sample processing cartridge; incubating the priming solution within the sample processing cartridge; dispensing one or more wash solutions into the inlet reservoir of the sample processing cartridge; transmitting solutions to a waste containment region of the sample processing cartridge; dispensing a cell suspension into the inlet reservoir of the sample processing cartridge and capturing cells, in single-cell format, within wells of the sample processing cartridge; dispensing a set of functionalized particles into the inlet reservoir of the sample processing cartridge and co-capturing the set of functionalized particles with the set of cells; incubating content of the wells of the sample processing cartridge; and picking up/releasing various tools involved with the substep(s). Steps S310''' and S315''' are preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(199) In more detail, performing a reverse transcription operation, at the sample processing cartridge, with lysis of the set of cells S320''' can include one or more of: dispensing one or more wash solutions into the inlet reservoir of the sample processing cartridge; transmitting solutions to a waste containment region of the sample processing cartridge; dispensing a particle-binding buffer into the inlet reservoir of the sample processing cartridge; dispensing a DTT solution into the inlet reservoir of the sample processing cartridge; dispensing a lysis solution into the inlet reservoir of the sample processing cartridge; displacing fluid above wells of the sample processing cartridge with an oil, thereby isolating contents of wells and preventing undesired material transfer across wells (e.g., as in U.S. application Ser. No. 16/564,375 filed 9 Sep. 2019, which is herein incorporated in its entirety by this reference); displacing the oil with air from the inlet reservoir; dispensing a particle-binding wash buffer into the inlet reservoir of the sample processing cartridge; dispensing a pre-RT reaction wash buffer into the inlet reservoir of the sample processing cartridge; dispensing a cDNA synthesis solution (e.g., SuperScript IV™) into the inlet reservoir of the sample processing cartridge; dispensing an RT cocktail into the inlet reservoir of the sample processing cartridge; incubating contents of the sample processing cartridge; performing incubation steps; and picking up/releasing various tools involved with the substep(s). Step S320''' is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(200) In more detail, retrieving the set of functionalized particles, with associated bound target

content, from the sample processing cartridge S325''' can include performing magnetic separation operations (e.g., as described above), using manual or automatic operations. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(201) In more detail, performing an exonuclease treatment operation with a volume of cell-derived content associated with the set of functionalized particles S330''' can include one or more of: mixing water and exonuclease buffer to produce an exonuclease solution having a desired concentration; dispensing the exonuclease solution, with functionalized particles into a first PCR container; dispensing an oil (e.g., mineral oil) into the first PCR container; thermocycling and incubating contents of the first PCR container; extracting a product of the first PCR container; performing a separation operation with the product of the first PCR container; discarding waste from the separation operation; performing incubation steps; and picking up/releasing various tools involved with the substep(s). Step S330''' is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(202) In more detail, performing a strand denaturing and second strand synthesis operation S335''' can include one or more of: transferring a hydroxide solution (e.g., sodium hydroxide solution) to a second magnetic separation container; mixing contents of the second magnetic separation container; activating a magnetic separation subsystem (e.g., set of magnets coupled to actuator, described above) in proximity to the second magnetic separation container, thereby separating functionalized magnetic particles toward the magnet(s); discarding waste material from the second magnetic separation container; dispensing a wash solution to the magnetic separation container; mixing a second strand synthesis primer enzyme within a process container in proximity to the second magnetic separation container; mixing the second strand synthesis primer enzyme with contents of the second magnetic separation container; dispensing product of the second magnetic separation container into a second PCR container; thermocycling contents of the second PCR container, with mixing; transferring product of the second PCR container to a third magnetic separation container; magnetically separating product of the third magnetic separation container from waste and discarding waste; transferring a wash solution to the third magnetic separation container; and picking up/releasing various tools involved with the substep(s). Step S335''' is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(203) In more detail, performing a cDNA synthesis operation S340''' can include one or more of: mixing a polymerase blend (e.g., Kapa Biosystems HiFi Hotstart Ready Mix™) within a cold storage volume; mixing the PCR master mix from the cold storage volume with contents of the third magnetic separation container; aliquoting contents of the third magnetic separation container into a third, fourth, fifth, and sixth PCR operation containers; aliquoting an oil (e.g., mineral oil) into the third, fourth, fifth, and sixth PCR operation containers; running a first PCR operation; and picking up/releasing various tools involved with the substep(s). Step S340''' is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(204) In more detail, performing a particle purification operation with PCR product of the cDNA amplification operation S345''' can include one or more of: dispensing product from the third, fourth, fifth, and sixth PCR operation containers into a fourth magnetic separation container; separating target content of the fourth magnetic separation container and transmitting target content into a fifth magnetic separation container; mixing and transferring PCR purification particles (e.g., AMPure beads XP™) into the fifth magnetic separation container; transferring product of the fifth magnetic separation container into a sixth separation container; and picking up/releasing various tools involved with the substep(s). Step S345''' is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can

be performed once, or repeated as recommended.

(205) In more detail, performing a set of antibody-derived tag purification and amplification operations with outputs of the particle purification operation **S350''** can include one or more of: mixing and transferring PCR purification particles (e.g., AMPure beads XP™) into the sixth magnetic separation container; discarding waste from the sixth magnetic separation container; diluting and mixing ethanol with nuclease-free water within a process container; transferring ethanol to the sixth magnetic separation container; discarding waste from the sixth magnetic separation container, upon performing magnetic separation; transferring target content of the sixth magnetic separation container into a seventh magnetic separation container; mixing PCR purification particles (e.g., AMPure beads XP™) into the seventh magnetic separation container; transferring ethanol to the seventh magnetic separation container; discarding waste from the seventh magnetic separation container, upon performing magnetic separation; dispensing water into the seventh magnetic separation container; transferring purified cDNA from the seventh magnetic separation container into a seventh PCR operation container; and picking up/releasing various tools involved with the substep(s). Step **S350'''** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(206) In more detail, performing a set of purification and amplification operations with outputs of the set of antibody-derived tag purification and amplification operations **S355''** can include one or more of: transferring PCR indexing primers into the seventh PCR operation container; transferring a polymerase blend into the seventh PCR operation container; mixing contents of the seventh PCR operation container; transferring an oil (e.g., mineral oil) into the seventh PCR operation container; initiating a second PCR operation; transferring PCR product from the seventh PCR operation container to an eighth magnetic separation container; transferring PCR purification particles (e.g., AMPure beads XP) into the eighth magnetic separation container; transferring ethanol to the eighth magnetic separation container; discarding waste from the eighth magnetic separation container, upon performing magnetic separation; dispensing water into the eighth magnetic separation container; transferring purified cDNA from the eighth magnetic separation container into a storage container; and picking up/releasing various tools involved with the substep(s). Step **S355'''** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(207) In more detail, performing a set of mRNA particle purification and amplification operations **S360''** can include one or more of: transferring nuclease-free water into the fifth magnetic separation container, with mixing and incubation; transferring product of the fifth magnetic separation container to a ninth magnetic separation container; mixing and transferring PCR purification particles (e.g., AMPure beads XP™) into the ninth magnetic separation container; transferring ethanol into the ninth magnetic separation container; removing waste from the ninth magnetic separation container, after incubation; mixing nuclease-free water with target content of the ninth magnetic separation container; mixing and incubating contents of the ninth magnetic separation container; transferring purified cDNA product from the ninth magnetic separation container to an eighth PCR operation container; transferring PCR master mix for mRNA amplification into the eighth PCR operation container; mixing contents of the eighth PCR operation container; transferring polymerase blend (e.g., Kapa Biosystems HiFi Hotstart Ready Mix) into the eighth PCR operation container; mixing contents of the eighth PCR operation container; transferring an oil (e.g., mineral oil) into the eighth PCR operation container; performing a third PCR operation within the eighth PCR operation container; transferring product of the third PCR operation from the eighth PCR operation container to a tenth magnetic separation container; transferring PCR purification particles into the tenth magnetic separation container; mixing contents of the tenth magnetic separation container; transferring ethanol to the tenth magnetic

separation container; discarding waste from the tenth magnetic separation container; transferring nuclease free water into the tenth magnetic separation container; mixing contents of the tenth magnetic separation container, with incubation; magnetically separating target mRNA-cDNA product from tenth magnetic separation container; transferring target mRNA-cDNA product from tenth magnetic separation container to cold storage; and picking up/releasing various tools involved with the substep(s). Step S360''' is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(208) In more detail, performing a run completion operation upon completion of the set of mRNA purification and amplification operations S365''' can include one or more of: returning the gantry to a home configuration; providing a notification that sample processing is complete; releasing the reagent cartridge and/or sample processing cartridge from the system for off-board storage; discarding system waste; performing a system cleaning operation; and transitioning the system to a deactivated (e.g., idle, off) state. Steps of S365''' can be implemented by the system automatically and/or by an operator. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(209) Example details of steps of the method 300''' are further described in TABLE 1 of Appendix C. In relation to steps of the method 300''', descriptions of ambient temperature and chilled reagents, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 2 of Appendix C. In relation to TABLE 2 of Appendix C, volumes of reagents can be scaled according to sizes of sample processing chips used and/or number of reactions run. In relation to magnetic separation operations, descriptions of apparatus and associated reagents used, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 3 of Appendix C. In relation to amplification (e.g., PCR) operations, descriptions of apparatus and associated reagents used, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 4 of Appendix C. In relation to fluid aspiration and delivery operations, descriptions of apparatus, as well as positions associated with an embodiment of the tool container described above are described in TABLE 5 of Appendix C. In relation to actuation of components for automation of protocol aspects, gantry arm and pipettor operations (e.g., in relation to apparatus coupling with disposables, apparatus uncoupling from disposables, fluid mixing, waste discarding, aspiration, delivery, aliquoting, etc.) are described in TABLE 6 of Appendix C. In relation to transitioning between modes for automation of protocol aspects, sample processing cartridge operations are described in TABLE 7 of Appendix C. In relation to transitioning between modes for automation of protocol aspects, heating and cooling subsystem operation modes are described in TABLE 8 of Appendix C. In relation to transitioning between modes for automation of protocol aspects, magnetic separation subsystem operations are described in TABLE 9 of Appendix C. In relation to amplification operations, PCR program details associated with the method 300''' are described in TABLE 10 of Appendix C.

(210) 3.4 Method—Example Workflow for Library Preparation

(211) As shown in FIG. 19, a variation of the method, configured for library preparation 300''' can include: performing a run preparation operation, wherein the run preparation operation configures a system for performing a library preparation protocol S305'''; performing a library preparation operation upon completion of the run preparation operation S310'''; performing a library purification operation with outputs of the library preparation operation S315'''; and performing a run completion operation upon completion of the library purification operation S360'''.

(212) In more detail, performing a run preparation operation S305''' can include substeps associated with one or more of: quantifying concentration of product (e.g., DNA concentration of product produced from a prior protocol; thawing out a reagent cartridge for library preparation, in a frozen storage state; processing storage volumes of the reagent cartridge (e.g., by vortexing, by

centrifugation etc.); diluting a sequencing adaptor (e.g., NEBNext Illumina™ adaptor) solution with a dilution buffer; returning the sequencing adaptor solution and previously removed storages to the reagent cartridge; performing operational checks of system subsystems (e.g., associated with the deck, associated with the gantry, associated with the base, etc.); returning the gantry to a home position; positioning the reagent cartridge at a deck of the system; removing one or more seals from the reagent cartridge and/or loading reagents onto the reagent cartridge; positioning a sample processing cartridge unit at a deck of the system; removing one or more seals from the tool container positioned at the deck; receiving an operator-loaded container (e.g., at a storage volume of the reagent cartridge) for performing the library preparation operation); initializing a heating and cooling subsystem (e.g., with an initial temperature set point); verifying proper positioning and states (e.g., in relation to expiration dates) of disposables for the protocol, upon scanning tags of disposables with a camera (e.g., machine vision camera); receiving sample identification information (e.g., from an operator); and initiating run of the sample. Steps of S305''' can be implemented by the system automatically and/or by an operator. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(213) In more detail, performing a library preparation operation upon completion of the run preparation operation S310''' can include sub-steps associated with one or more of: transferring diluted cDNA from the operator-loaded tube into a first cold storage container containing buffer; transferring contents of the first cold storage tube to a second cold storage container; incubating contents of the second cold storage container; transferring a cDNA mixture from a second ambient storage container into a first PCR operation container; fragmenting content of the first PCR operation container upon performing thermocycling operations at the first PCR operation container; transferring fragmented DNA from the first PCR operation container to a fourth cold storage container, with mixing; transferring contents of the fourth cold storage container to a fifth cold storage container with mixing; transferring diluted adaptor from a third cold storage container to the fifth cold storage container with mixing and incubation; transferring contents of the fifth cold storage container to a second PCR operation container with incubation; transferring contents of the second PCR operation container to a sixth cold storage container with mixing; transferring contents of the sixth cold storage container to the second PCR operation container with incubation; transferring contents of the second sixth cold storage container to a second magnetic separation container; transferring PCR purification particles (e.g., AMPure beads XP) to the second magnetic separation container with mixing; discarding waste from the second magnetic separation container; transferring ethanol to the second magnetic separation container; discarding waste from the second magnetic separation container; transferring TE buffer to the second magnetic separation container with incubation and magnetic separation; transferring purified cDNA from the second magnetic separation container to a third PCR operation container; transferring indexing PCR master mix to the third PCR operation container with mixing; performing a fourth PCR operation; performing mixing steps; performing incubation steps; and picking up/releasing various tools involved with the substep(s). Step S310''' is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(214) In more detail, performing a library purification operation with outputs of the library preparation operation S315''' can include sub-steps associated with one or more of: transferring product of the fourth PCR operation from the third PCR operation container to a third magnetic separation container with mixing; transferring PCR purification particles (e.g., AMPure Beads XP) to the third magnetic separation container with mixing, incubation, and magnetic separation; transferring ethanol to the third magnetic separation container with incubation; removing waste from the third magnetic separation container; transferring nuclease-free water to the third magnetic separation container with mixing, incubation, and magnetic separation; transferring contents of the third magnetic separation container to a fourth magnetic separation container with mixing;

transferring PCR purification particles (e.g., AMPure Beads XP) to the fourth magnetic separation container with mixing, incubation, and magnetic separation; removing waste from the fourth magnetic separation container; transferring ethanol to the fourth magnetic separation container with incubation; discarding waste from the fourth magnetic separation container; repeating steps for further purification, with transfer of target material from the fourth magnetic separation container to a fifth magnetic separation container, to a sixth magnetic separation container; transferring nuclease-free water to the sixth magnetic separation container with incubation and magnetic separation; transferring purified cDNA for library construction to an eighth cold storage container; performing mixing steps; performing incubation steps; and picking up/releasing various tools involved with the sub-step(s). Step **S315** is preferably performed automatically by the system but can alternatively be performed in another suitable manner. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(215) In more detail, performing a run completion operation upon completion of the library purification operation **S360** can include one or more of: returning the gantry to a home configuration; providing a notification that sample processing is complete; releasing the reagent cartridge and/or sample processing cartridge from the system for off-board storage; discarding system waste; performing a system cleaning operation; and transitioning the system to a deactivated (e.g., idle, off) state. Steps of **S360** can be implemented by the system automatically and/or by an operator. Furthermore, various sub-steps can be performed once, or repeated as recommended.

(216) Example details of steps of the method **300** are further described in TABLE 1 of Appendix D. In relation to steps of the method **300**, descriptions of ambient temperature and chilled reagents, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 2 of Appendix D. In relation to TABLE 2 of Appendix D, volumes of reagents can be scaled according to sizes of sample processing chips used and/or number of reactions run. In relation to magnetic separation operations, descriptions of apparatus and associated reagents used, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 3 of Appendix D. In relation to amplification (e.g., PCR) operations, descriptions of apparatus and associated reagents used, as well as positions associated with storage volumes of an embodiment of the reagent cartridge described above are described in TABLE 4 of Appendix D. In relation to fluid aspiration and delivery operations, descriptions of apparatus, as well as positions associated with an embodiment of the tool container described above are described in TABLE 5 of Appendix D. In relation to actuation of components for automation of protocol aspects, gantry arm and pipettor operations (e.g., in relation to apparatus coupling with disposables, apparatus uncoupling from disposables, fluid mixing, waste discarding, aspiration, delivery, aliquoting, etc.) are described in TABLE 6 of Appendix D. In relation to transitioning between modes for automation of protocol aspects, sample processing cartridge operations are described in TABLE 7 of Appendix D. In relation to transitioning between modes for automation of protocol aspects, heating and cooling subsystem operation modes are described in TABLE 8 of Appendix D. In relation to transitioning between modes for automation of protocol aspects, magnetic separation subsystem operations are described in TABLE 9 of Appendix D. In relation to amplification operations, PCR program details associated with the method **300** are described in TABLE 10 of Appendix D.

(217) The system embodiment(s) can, however, be configured to implement other workflows including variations of those described, and/or other workflows.

(218) 4. Conclusion

(219) The FIGS. illustrate the architecture, functionality and operation of possible implementations of systems, methods and computer program products according to preferred embodiments, example configurations, and variations thereof. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable

instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block can occur out of the order noted in the FIGS. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(220) As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

Claims

1. A system for processing a sample, the system comprising: a reagent cartridge; a deck configured to support and position: a sample processing cartridge comprising a set of wells, a lid configured to cover an access region over the set of wells of the sample processing cartridge, the lid comprising an open mode in which the access region is uncovered and a closed mode in which the access region is covered, the lid further comprising a releasing element; and a gantry coupled to the deck and comprising: a set of tracks defining paths of movement along a set of axes, within a three dimensional volume bound by a first side of the deck, for a pipette interface; a heating and cooling subsystem comprising a thermal body configured to thermally interface with the sample processing cartridge; and a lid opening tool configured to transition the lid between the open mode and the closed mode, with retention of the sample processing cartridge thermally interfaced with the heating and cooling subsystem, wherein the lid-opening tool comprises a linking element configured to engage the releasing element on the lid.
2. The system of claim 1, wherein the sample comprises nucleic acids.
3. The system of claim 1, wherein the sample comprises a cell suspension.
4. The system of claim 1, wherein the reagent cartridge comprises materials for performing polymerase chain reaction (PCR) operations with the sample.
5. The system of claim 1, wherein the reagent cartridge comprises materials for performing library preparation operations with the sample.
6. The system of claim 1, wherein the reagent cartridge comprises materials for performing reverse transcription and cDNA synthesis.
7. The system of claim 1, wherein the lid comprises a gasket configured to seal the access region of the sample processing cartridge in the closed position.
8. The system of claim 1, wherein the lid comprises a latch configured to couple with the sample processing cartridge.
9. The system of claim 1, wherein further comprising a camera coupled with a pipettor coupled to the gantry, the camera having a field of view including objects that the pipettor is aligned with.
10. The system of claim 1, further comprising a separation subsystem comprising a set of magnets coupled to a magnet actuator configured to transition the set of magnets between a retracted state and an extended state, wherein in the extended state, the set of magnets passes into an opening of the deck.
11. A method comprising: retaining a sample processing cartridge at a deck; wherein the sample processing cartridge comprises: a set of wells, and a lid configured to cover an access region over the set of wells of the sample processing cartridge, the lid comprising an open position in which the access region is uncovered and a closed position in which the access region is covered, the lid further comprising a releasing element; transferring a sample to the set of wells of the sample processing cartridge; performing a set of operations in coordination with transferring a set of

reagents from a reagent cartridge to the set of wells of the sample processing cartridge, wherein the set of operations comprises: priming the sample processing cartridge upon dispensing a priming solution into the sample processing cartridge; performing a reverse transcription operation, at the set of wells of the sample processing cartridge, on nucleic acids of the sample; performing a strand denaturing and second strand synthesis operation upon performance of the reverse transcription operation; transitioning, using a lid-opening tool the lid between the open position and the closed position, with retention of the sample processing cartridge at the deck, wherein the lid-opening tool comprises a linking element configured to engage the releasing element on the lid; and performing a cDNA amplification operation upon performance of the strand denaturing and second strand synthesis operation.

12. The method of claim 11, further comprising binding released mRNA content of the sample to barcoded functionalized particles within the set of wells.

13. The method of claim 11, wherein transitioning the lid between the open position and the closed position comprises applying a force to the lid, thereby isolating the access region of the sample processing cartridge.

14. The method of claim 11, wherein the deck is coupled to a gantry comprising a set of tracks defining paths of movement along a set of axes, the method further comprising: with a camera coupled with a pipettor coupled to the gantry, scanning a tag of the sample processing cartridge.

15. The method of claim 14, further comprising guiding movement of components of the gantry, to within 100 micron precision, with feedback using a camera image generated by the camera.

16. The method of claim 11, further comprising performing a set of antibody-derived tag purification and amplification operations with outputs of the cDNA amplification operation, for performance of a CITE-seq single cell multiomics protocol.

17. The method of claim 11, further comprising performing a library preparation operation with material of the sample, wherein the reagent cartridge contains a set of reagents for the library preparation operation, and wherein performing the library preparation operation comprises fragmenting and amplifying nucleic acid content of the sample.

18. The method of claim 17, wherein performing the library preparation operation comprises determining a concentration of a DNA component of the sample; based upon the concentration performing a dilution operation upon the sample; fragmenting cDNA material of the sample; and amplifying fragmented cDNA of the sample.

19. The method of claim 17, further comprising performing a library purification operation with outputs of the library preparation operation upon combining material from the library preparation operation with PCR purification particles, by way of a pipettor coupled to an actuator of a gantry coupled to the deck.

20. The method of claim 11, further comprising performing a separation operation upon the sample, wherein performing the separation operation comprises actuating a magnet toward magnetic separation reservoir at the deck.
